Starting Up AI

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Abstract

Using comprehensive administrative data on business applications over the period 2004- 2023, we study business applications (ideas) and the resulting startups that aim to develop AI technologies or produce goods or services that use, integrate, or rely on AI. The annual number of new AI-related business applications is stable between 2004 and 2011, but begins to rise in 2012 with further increases from 2016 onward into the Covid-19 pandemic and beyond, with a large, discrete jump in 2023. The distribution of these applications is highly uneven across states and sectors. AI business applications have a higher likelihood of becoming employer startups compared to other applications. Moreover, businesses originating from these applications exhibit higher revenue, average wage, and labor share, but similar labor productivity and lower survival rate, compared to other businesses. While it is still early in the diffusion of AI, the rapid rise in AI business applications, combined with the better performance of resulting businesses in several key outcomes, suggests a growing contribution from AI-related business formation to business dynamism.

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1 Introduction

Artificial Intelligence (AI) is a collection of technologies that can perform tasks that require human intelligence, such as visual perception, speech recognition, and decision-making. For several years now, industry analysts, academics, and the popular press have been hypothesizing about how AI will transform the future of work and productivity (see, e.g., Baily et al. (2023); Acemoglu and Restrepo (2020); Agrawal et al. (2019)). But how fast is AI diffusing among U.S. firms? Evidence from recent U.S. Census Bureau surveys suggests that the rate of AI use by firms in producing good and services remains relatively low, and adoption is concentrated mostly among large firms.[1] Nevertheless, there are now signs that AI diffusion is accelerating, as the recent estimates from the Census Bureau's Business Trends and Outlook Survey (BTOS) indicate that AI use rate among firms is nearly 5.5% as of late 2023, up from 3.2% for the period 2016-2018 based on the Annual Business Survey (ABS).[2] Consistent with this acceleration, forward-looking questions in BTOS on expected future AI use by businesses indicate that the use rate may reach as high as 8.5% by Fall 2024.[3]

Why is AI adoption apparently low? Aside from the difficulties of measuring AI use in surveys and the resulting potential underestimation of AI adoption rate, there are many reasons.[4] AI is still a nascent technology with growing applications to a diverse set of business problems. These applications can only be developed over time with specific technological breakthroughs in AI and related technologies – a recent example is the development of generative AI and its applications.[5] Moreover, many firms lack scale, complementary technologies, and relevant infrastructure, such as high computing power and vast amounts of data in digital form, to benefit from some AI applications.[6] Thus, many small businesses, which represent the bulk of the firm population, do not adopt AI, and most may not yet find it readily applicable to their core business – a pattern that may be changing with the advent of generative AI and its applications to business problems. The limited applicability and the high cost of AI are indeed often cited as key barriers to wider

[1] See, e.g., McElheran et al. (2024); Zolas et al. (2020); Acemoglu et al. (2022).

[2] See Acemoglu et al. (2022), Bonney et al. (2024), Breaux and Dinlersoz (2023), and https://www.census.gov/hfp/btos/data. The AI use rate from BTOS is adjusted to remove "don't know" answers to make it comparable to the ABS adoption rate. The unadjusted BTOS use rate is about 5%.

[3] For bi-weekly estimates of current and expected future AI use rates, visit BTOS - Data (census.gov). This figure reflects the adjustment to exclude "don't know" answers to the future adoption question. The unadjusted rate is about 7%.

[4] Difficulties in measuring AI can lead to measurement error in firm-level surveys. Many recent surveys, including those by the U.S. Census Bureau, focus on measuring the use of AI specifically as part of a production process for goods or services, or AI as part of a product or service provided by the firm. Many firms, or their employees, may be likely to use AI only incidentally or insignificantly but not report such use, as AI may not be a part of their core processes or products. Others may rely on AI embedded in systems or services supplied by third-parties, in which case they are unknowing users of AI.

[5] Generative AI is a form of artificial intelligence that is capable of generating text, images, or other types of data based on generative models, often in response to a prompt in a user interface.

[6] See Zolas et al. (2020) for discussion on digital hierarchies.

adoption and use among firms.[7]

AI diffusion is fueled by two margins: the entry of new businesses using or providing AI, and the adoption or provision of AI by existing businesses. Existing businesses can benefit from AI only to the extent they can change the way work is organized to utilize AI effectively. For example, productivity improvements from AI may materialize only when workers are retrained and existing processes and systems are upgraded. Recent data from the BTOS indicate that as many as 20% of AI-using firms trained existing staff to use AI or developed new workflows.[8] This need for organizational changes is one reason why it may be sometimes easier for new, early-stage businesses to take advantage of emerging technologies, including AI, as they can readily integrate these technologies with business processes early on or even design the business around a new technology.[9] The number of early-stage or nascent businesses planning to use or develop AI may thus be increasing fast, along with a rise in new AI-using startups emerging from this activity. Less is known about this margin of the diffusion of AI among U.S. firms.

In this paper, we provide new evidence on the prevalence of AI-related business ideas among potential entrepreneurs who seek to start new businesses. Instead of surveys, we rely on comprehensive administrative micro data of business applications from the U.S. Census Bureau's Business Formation Statistics (BFS) to identify AI-related business applications (henceforth, *AI business applications*) over the period 2004-2023, and employer startups originating from these applications. The BFS micro data is based on applications for Employer Identification Numbers (EINs) filed with the IRS using Form SS-4, and covers nearly the entire set of economically significant business initiation activity. Critically, each application contains information on the nature of the planned business. Using text analysis on an applicant's reported main business line, business name, and trade name in conjunction, we are able to identify business ideas for developing or providing AI technologies and applications, as well as those that aim to use AI in providing goods or services. Our approach, based on administrative data, is complementary to surveys that measure AI adoption and use, and particularly informative on early-stage business activity not typically captured in surveys. Because business applications are received and monitored weekly on a continuous basis by the Census Bureau's BFS program, our analysis is also promising in terms of near real-time and high-frequency measurement of AI-related business formation activity going forward.

We find that new AI business applications arrive at a largely stable rate annually for the period 2005-2012. The annual number of AI business applications begin to increase starting in 2012. Beginning around 2016, applications exhibit a faster rise that continues into the Covid-19

[7] See Acemoglu et al. (2022) and Bonney et al. (2024)

[8] See Bonney et al. (2024).

[9] This is consistent with the fact that young firms are more likely adopters of new and advanced technologies (including AI), conditional on other characteristics – see, e.g., Zolas et al. (2020); Acemoglu et al. (2022).

Pandemic and beyond. In particular, we observe a large, discrete jump in application activity in 2023, potentially attributable to the recent advances in generative AI. We provide several pieces of external evidence in support of this observed time-path and how it relates to key events in the development of AI technologies and applications over the last two decades.

We document that the annual number of employer startups originating from AI business applications have also increased rapidly in recent years, driven by the rising number of applications and the fact that AI business applications have a higher than average rate of transitioning to employer businesses. Nevertheless, it may take some time for this accelerating AI startup activity to make a big dent in the overall AI adoption rate in the firm population. This is in part because the current levels of annual AI business applications ad the resulting startups are still small relative to the number of existing firms in the U.S. Furthermore, the rates of adoption and de-adoption by existing firms are also important for the overall adoption rate.

In terms of the distribution of the AI business application activity, we observe a high degree of concentration across sectors and states over the analysis period: nearly 56% of all applications are in Professional, Scientific, and Technical Services (NAICS 54), and about 20% originate from California – consistent with the patterns of high sectoral and geographic concentration observed in surveys, particularly in the ABS and BTOS.[10]

We also examine the early characteristics of AI business applications and compare them with those of other applications. Each business application is accompanied by a set of characteristics submitted by the applicant, such as the reason for application, intent to hire employees, legal form of organization, previous EIN application, plans to make wage payments, and expected initial employment. Many of these characteristics may provide, mostly indirectly, information on the underlying viability of the business idea. We show that AI business applications have initial characteristics associated with a higher potential to become an employer business. They also anticipate higher initial startup scale (employment) compared to other business applications. Conditional on application characteristics, employer businesses that form from AI business applications tend to be larger in terms of employment and revenue, and have higher payroll, average pay per employee, and labor share in their early life-cycles, compared to employer businesses forming from other applications. However, they exhibit similar labor productivity and lower survival rate.

In related research using early-stage startups, McElheran et al. (2024) find that conditional on survival AI-using young firms tend to exhibit faster revenue growth than others, consistent with our findings that AI startups perform better in some key outcomes. Our findings are also consistent with the evidence from the BTOS that AI-using businesses generally have better employment and sales performance (Bonney et al. (2024)), and the evidence on the positive relationship between

[10]See Zolas et al. (2020); Acemoglu et al. (2022); McElheran et al. (2024); Bonney et al. (2024).

firm performance and AI use based on AI-related job postings and worker resumes (Babina et al. (2022)).

We emphasize that our findings do not necessarily indicate a causal link between AI use and business performance; our goal is to describe the key features of emerging AI business applications and ensuing businesses. Moreover, the patterns we document pertain to the early stages of the diffusion of AI and can change as the diffusion progresses. Nevertheless, the recent acceleration in AI business applications and their higher rates of transition to employer businesses, along with the better performance of AI employer businesses in a number of key outcomes, suggest a currently small but increasing contribution of new AI business activity to business dynamism and the economy – especially in light of the fact that business applications with high likelihood of becoming employers are early indicators of future employer business formation and employment growth (see Asturias et al. (2021, 2023)). Thus, our findings can be viewed as an early indicator of the rise of AI use among firms and an initial documentation of the emerging relationships between AI use and business characteristics. Our back of the envelope calculations suggest that aggregate adoption rates of AI will increase from 5.5% in 2023 to 9-10% in 2027.

The paper proceeds as follows. We lay out our conceptual framework in Section 2. Section 3 describes the data. Section 4 presents our findings. A brief summary of the results and a discussion are provided in section 5. Section 6 concludes.

2 Conceptual Framework

To conceptualize our measurement of AI business applications and formations, we consider a simple framework. In any period (year) t, there are N_t employer businesses in an economy, N_t^a of which use or provide AI – collectively referred to as AI businesses. Define the share of AI businesses as

$$a_t = \frac{N_t^a}{N_t}. \tag{1}$$

In the beginning of period $t+1$, businesses that have not yet used or provided AI start doing so with a rate r_{t+1}, and a fraction d_{t+1} of businesses that have used or provided AI in period t cease to do so. In addition, E_{t+1}^a new AI businesses enter during period $t+1$, while X_{t+1}^a of the existing AI businesses exit at the end of the period. As a result, the stock of AI businesses evolves according to

$$N_{t+1}^a = N_t^a(1-d_{t+1}) + r_{t+1}(N_t - N_t^a) + E_{t+1}^a - X_{t+1}^a. \tag{2}$$

The share of AI businesses at the end of period $t+1$ is then

$$a_{t+1} = \frac{N_t^a(1-d_{t+1}) + r_{t+1}(N_t - N_t^a) + E_{t+1}^a - X_{t+1}^a}{N_{t+1}}. \tag{3}$$

Dividing the numerator and the denominator through by N_t yields

$$a_{t+1} = \frac{a_t(1-d_{t+1}) + r_{t+1}(1-a_t) + (\varepsilon_{t+1}^a - \chi_{t+1}^a)}{1+g_{t+1}}, \tag{4}$$

where $g_{t+1} = (N_{t+1}/N_t) - 1$ is the growth rate of the total number of businesses in the economy and

$$\varepsilon_{t+1}^a = \frac{E_{t+1}^a}{N_t}, \;\; \chi_{t+1}^a = \frac{X_{t+1}^a}{N_t} \tag{5}$$

are the entry and exit rates of AI businesses in period $t+1$, measured relative to the stock of all businesses at the end of period t.

Based on (4), the adoption rate is fueled by two key margins: the share of existing businesses that adopt, $r_{t+1}(1-a_t)$, and the entry rate of new AI businesses, ε_{t+1}^a. Note that the entry rate plays a relatively more important role if the adoption rate by existing businesses (r_{t+1}) is low. The share of AI businesses increases from one period to another ($a_{t+1} > a_t$) if

$$\varepsilon_{t+1}^a > a_t g_{t+1} + d_{t+1}a_t + \chi_{t+1}^a - r_{t+1}(1-a_t). \tag{6}$$

In other words, the rate of entry needs to be sufficiently high to surpass the current adoption rate adjusted by the overall growth in firm population, and overcome the de-adoption and exit rate of existing AI firms – net of the rate of increase in the fraction of existing firms that use AI.

The focus of our empirical analysis is on the entry rate

$$\varepsilon_{t+1}^a = E_{t+1}^a/N_t = (I_{t+1}^a \tau_{t+1})/N_t = \rho_{t+1}\tau_{t+1}, \tag{7}$$

where I_{t+1}^a is the stock of AI business applications (or ideas) that can potentially transition into employer businesses at the beginning of period $t+1$, and τ_{t+1} is the transition rate of each such idea to an actual startup in period $t+1$. The entry rate is higher, the higher the ratio of AI business applications to the number of existing businesses, ρ_{t+1}, or the higher the transition rate of ideas, τ_{t+1}.

The stock of AI business ideas at risk of transitioning can be written as $I_{t+1}^a = (1-\tau_t)I_t^a + F_{t+1}^a$, where F_{t+1}^a is the flow of new AI business applications in period $t+1$. The empirical analysis first provides some estimates of the annual flow of AI business applications, F_t^a, their transition rate, τ_t, and the resulting number of startups. In addition, we explore the characteristics of AI business

applications to understand how they differ from other business applications. Finally, we study the performance of employer businesses that originate from AI business applications in comparison to those that originate from other business applications. Census Bureau surveys generated estimates of a_t for some of the recent years, as well as preliminary estimates of d_{t+1} and r_{t+1}.[11] Using some of these estimates and our findings, we also provide estimates of the diffusion of AI among firms in the near term based on the conceptual framework.

3 Data

3.1 Business Formation Statistics

The core analysis relies on micro data underlying the U.S. Census Bureau's Business Formations Statistics (BFS). Next, we describe key aspects of BFS relevant for our analysis – see also Bayard et al. (2018) and Dinlersoz et al. (2023) for background on BFS micro data.

3.1.1 Business Applications

The BFS provides high-frequency statistics on early-stage entrepreneurial activity based on applications for employer identification numbers (EINs).[12] Applications for new EINs are received by the Census Bureau on a weekly basis from the IRS. Each EIN application, made using IRS Form SS-4, contains information on a variety of business and owner characteristics and is assigned a detailed industry codes (6-digit NAICS) and a geocode. Based on this information, the BFS identifies a core set of EIN applications that are primarily made with an intent to start a business, by filtering out applications made for other reasons (e.g., for trusts, estates, and other financial filings), applications from public entities, applications originating outside of the U.S., and applications related to certain agricultural, financial, or private household activity.[13]

Employer businesses are required to have an EIN to file payroll taxes. Similarly, all businesses (employer or nonemployer) must file for an EIN to form a partnership or a corporation. The BFS data does not cover some business types that can operate without an EIN – mainly sole proprietor nonemployers. However, many of these businesses still need and use an EIN to establish an official business bank account, establish a more formal business identity, or to conform to certain IRS regulations (Fairlie et al. (2023)). While there are many sole proprietor nonemployers without

[11] See Zolas et al. (2020); Acemoglu et al. (2022); McElheran et al. (2024); Bonney et al. (2024) for recent estimates of a_t.

[12] For public-use BFS data products, see the BFS website.

[13] The details of the micro data and the construction of the BFS are in Bayard et al. (2018), including the exact set of filters used to exclude applications.

EINs, they account for a small fraction of aggregate economic activity.[14] Thus, the BFS micro data on applications provides nearly full coverage of all economically significant business initiations.

The information on the application form includes the name and address of the applicant and business, application week, business start date, reason for application (hiring, starting a new business, purchasing an ongoing business, banking, etc.), type of business entity, previous application for an EIN, industry, main line of business, planned date for initial wage payments, and the highest number of employees expected in the first 12 months following application (the latter is available after 2017). Some of these characteristics provide information on the underlying business idea and are key predictors of the likelihood of transitioning to an employer business. Bayard et al. (2018) and the BFS program use these characteristics in probit or linear probability models (LPM) to predict the likelihood of transition and generate projected business formations for recent time periods for which actual formations are not yet available. This prior work finds that applications are more likely to transition when they indicate a planned wage payment date, are filed by a corporate entity, are made with a goal to hire or start a new business (versus other reasons, such as banking or conformity to IRS regulations), or are based in certain industries, such as retail and certain professional services.

Our focus is on business applications related to AI. Just like the rest of the applications, these applications can be intended for an employer or a nonemployer business – e.g., AI code developers, researchers, or consultants. However, for job creation and contribution to overall economic activity, applications that are more likely to transition to employer businesses are critical.[15] Using application characteristics, the BFS program identifies a subset of all applications as "high-propensity business applications (HBA)". These are applications that (i) are made by a corporate entity, (ii) indicate hiring employees, (iii) provide a (planned) first wages-paid date, or (iv) have a NAICS industry code in specific sectors and industries – accommodation and food services (NAICS 72) or in portions of construction (NAICS 237, 238), manufacturing (NAICS 312, 321, 322, 332), retail (NAICS 44, 452), professional, scientific, and technical services (NAICS 5411, 5413), educational services (NAICS 6111), and health care (NAICS 621, 623).[16] High-propensity applications transition to employer status with a much higher likelihood (23.9%) than non-high propensity (5.3%) over the sample period. We therefore explore the extent to which AI business applications are "high-propensity", and whether they disproportionately possess certain features of "high-propensity" applications such as indicating planned wage payments or originating from corporations.

While business applications are available at a monthly or weekly frequency, annual data is

[14] Furthermore, Davis et al. (2009) find that nonemployers with an EIN (including sole proprietors) are on average much larger in terms of revenue than those without an EIN.

[15] Asturias et al. (2021, 2023) show that applications that are more likely to transition to employers are especially early indicators of aggregate economic activity.

[16] See Business Formation Statistics - Methodology (census.gov) and Bayard et al. (2018).

sufficient for the purposes of understanding the recent broad evolution of AI business initiation activity. The annual frequency also avoids the complications of higher noise and seasonality in the monthly and weekly data.[17] For our analysis, we mainly focus on the period 2004q3-2023m10 – the 2004 data is only available starting in the third quarter, and the 2023 data contains all months until November. As we discuss below, this period covers years both before and after key technological developments and milestones relevant for the recent surge in AI business ideas.

3.1.2 Business formations

BFS combines the confidential micro data underlying the EIN applications with the Longitudinal Business Database (LBD) – the longitudinal version of the Census Bureau's Business Register. The LBD contains firm and establishment-level information on age, location, industry, number of employees, quarterly payroll, and EIN for employer businesses. BFS uses the EIN to match business applications to the LBD to identify the incidence and timing of transitions to new employer businesses.

In documenting employer business formations (i.e., transitions or startups) from AI business applications, we focus on entirely new employer businesses (i.e. "age zero" firms in the LBD) that form within 8 quarters of the application date – nearly 90% percent of transitions to employer businesses occur within eight quarters of the application.[18] We define the transition rate as the ratio of the number of transitions within 8 quarters to the number of business applications these transitions originate from. Because the latest vintage of the LBD covers years up to 2021, we do not observe 8-quarter window formations from applications made after 2019. Thus, for the period 2020-2023 we use projected (or predicted) formations the BFS provides based on a linear probability model (LPM) that leverages the rich set of application characteristics as predictors.[19]

We also follow AI business applications after transition to explore whether AI startups systematically differ in business outcomes from other startups. In tracking post-transition performance of startups, we use all new employer businesses originating from applications without the 8-quarter window restriction – thus including all employer business formations until the end of 2021. As we detail below, in this analysis we control for initial application characteristics as well as the startup life-cycle stage (summarized by size and age).

Overall, the micro data contains more than 60 million applications and nearly 8 million employer startups since 2004. We use this rich data to identify AI business applications and startups based on a methodology which we describe next.

[17] However, in the future we plan to use monthly data (with seasonal adjustment) to track AI business applications over time at a high frequency.

[18] See Figure 2 in Dinlersoz et al. (2023).

[19] See Bayard et al. (2018) for a description of the LPM model that the BFS program uses to track the evolution of predicted formations for periods for which the actuals are not yet available.

3.2 Identifying AI business applications

To identify AI business ideas or applications, we rely on three key variables in the BFS micro data obtained directly from EIN applications: main business line (principal line of merchandise), business name, and trade name. These variables contain the text written-in by an applicant, and provide information on the planned main product or service, and the name of the business. We search these three text fields in conjunction to isolate cases that contain key words related to AI technologies and applications.[20] Importantly, we do not restrict our search to certain industries with more intense reliance on AI (e.g., high-tech industries or professional services), since we want to capture AI business initiations across all sectors as they emerge.

As a first step, we form a comprehensive dictionary of AI-related keywords based on the accumulated AI literature and practice, and input from experts (Emerging Technology Fellows from "xD" at the Census Bureau).[21] This large dictionary spans several AI technologies and their applications, broadly or narrowly defined, such as artificial intelligence, machine learning, deep learning, computer/machine vision, image processing, speech/voice recognition, neural networks, large language models, virtual agents, generative AI, etc. It also includes several keywords on business implementations of AI technologies, such as robotics process automation, self-driving cars, AI art, etc. We supplement this dictionary with a list of additional keywords related to AI technologies and their applications provided by ChatGPT. We also add variations of several AI-related keywords to allow for slight deviations or spelling mistakes, common abbreviations, and closely related phrases and synonyms.

In the second step, we use the dictionary to identify business applications related to AI by searching the three text fields and finding matches to the entries in the dictionary. Finally, we perform extensive filtering to exclude those applications that are not aimed at AI businesses. In filtering, we use all the information available in main business line, business name, and trade name in relation to each other to remove "false positives". If the business name or trade name for an application contains an AI keyword, we cross-check the main business line information to ensure that the cases unrelated to AI businesses are removed. For example, a business application is excluded if it has an AI-related keyword in its business name or trade name, but indicates a main business line unrelated to AI technologies or applications – such as in the case of *fictitious* business names "ABC Superior AI Services" and "Master Chef Ai Chung's Garden" with main business lines "artificial insemination for cattle" and "Chinese restaurant", respectively.

We also remove the types of applications made for purposes not related to core AI business

[20] Colaiacovo et al. (2023) use a similar approach, albeit based on firm names only, to identify whether a business registration in a state is associated with a particular technology.

[21] In particular, we thank Tomo Lazovich and Mike Walton for their input. To learn more about xD at Census, visit xD - U.S. Census Bureau.

activity, such as hobbies, clubs, trade associations/shows, conventions/conferences, investment vehicles, and holding companies. However, the time trends in AI business applications, and the distributions of the applications across states and sectors, are highly robust to various filtering criteria we experimented with. Overall, our filtering results in a set of business applications that are closely related to the provision or use of AI technologies.

Table 1 shows the number of AI business applications we identify, along with the filtered number of applications. We break down the applications identified by each field of search (main business line, business name, trade name) – some applications may be in the intersection of two or all fields. A large number of AI business applications are identified based on main business line and business name, and a smaller subset based on trade name – mainly because a business is not required to have a trade name, unlike a business name and main line of business, which are available for almost all applications in our data. In some of the analysis to follow, we consider separately AI business applications based on main business line and the complement set that consists of AI business applications identified solely based on business/trade name to document any differences between the two groups. Overall, we end up with 21,731 AI business applications after filtering, 11,545 (53%) of which are identified based on main business line (and possibly on business/trade name), and the remaining 10,186 (47%) based on business/trade name only. Table 1 also shows that, perhaps not surprisingly, AI business applications constitute a very small fraction (0.035%) of all business applications in our database.

AI business applications we identify include both the providers and users of AI technologies. Applications identified based on main business line tend to contain more AI providers and include those that report a main business line aimed to develop AI technologies, provide AI code/software, build or integrate AI systems, or offer AI consulting services - some of these are likely nonemployers. The ones identified based on business/trade name only are primarily users of AI technologies in their planned products or processes. However, both groups are ultimately a mix of both providers and users, and potential employers and nonemployers. For the current analysis, we do not attempt to classify applications into users or providers.[22] Overall, we view AI business applications in its entirety as a "business ecosystem" or "industry" for AI-related activity – including users, innovators, developers, coders, consultants, and suppliers of AI.

[22]The information provided in the text fields we utilize is not always detailed enough to make that classification with precision. There are additional, non-text application characteristics that can be useful for classification – we show below that some of these characteristics are strongly associated with AI business applications. Future work will explore machine learning approaches to potentially improve the identification of AI business applications and their types – see, e.g., Miric et al. (2023) for a recent application of machine learning methods to the identification of AI patents.

3.3 Selection

Our methodology for identifying AI business applications depends on potential entrepreneurs' propensity to reveal the nature of their planned business by using AI-related keywords in their main business line, business name, or trade name. AI business applicants who see only a supporting or less significant role for AI in their potential businesses may not reveal such information. Thus, our approach may be better suited for identifying applications that aim to develop or provide AI technologies or use AI in a significant part of the planned processes, products, or services. It may also be the case that savvy AI entrepreneurs use AI-related keywords in their business names or descriptions, while others do not, despite having an AI-related business.[23] Our classification is therefore subject to potential selection issues associated with reporting AI-related terms.[24] Similar selection issues are also encountered in surveys aiming to measure AI use by firms: non-response that is not random, incidental or insignificant use of AI that gets reported as no use, and unknowing use of AI that is not reported.

Because we do not necessarily identify all underlying AI-related business activity based on the revealed textual information alone, the number of applications we end up with likely underestimate the underlying extent of certain AI-related business activity.[25] While we do not necessarily capture the exact underlying *level* of AI business applications, our approach is likely more informative on the *growth* in AI business applications. Below we rely on several approaches to externally validate the growth trends we observe.

We do not know the exact nature of the selection into reporting AI-related keywords with respect to the unobserved true set of AI business applications. Nevertheless, we can compare the characteristics and outcomes of AI business applications we identify with those of all other business applications. As we detail below, we find that they tend to have higher transition likelihood to employer businesses, and exhibit better post-transition outcomes in certain dimensions. We also show that some of these differences are driven by observable initial features of AI business applications.

[23] Some entrepreneurs may also be riding the "AI wave" by using catchy AI-related phrases to name businesses that use AI only incidentally. However, note that our results hold when we use the information in main business line only. Therefore, it is unlikely that the trends are driven entirely by a desire to use AI-related phrases in business name.

[24] This selection is a more general challenge for the identification of any specific type of business activity in the BFS micro data, and not unique to the identification of AI business applications. See also Colaiacovo et al. (2023) for a discussion of similar issues in identifying businesses related to a given technology in state business registrations.

[25] Even though we eliminate false positives, there remains the possibility, although less likely, that some AI keywords are used by applicants who do not intend to use AI in their planned business.

4 Findings

We first describe the evolution of AI business applications and actual employer businesses that form from these applications annually over the period 2005-2023. We then summarize the patterns of applications across states and sectors.

4.1 Time-path of AI business applications and formations

Figure 1 shows the annual number of AI business applications over time. Applications come in at a steady low annual rate over the period 2005-2011. The annual number of applications starts to increase around 2012. The second change in the series occurs around 2016, after which there is further acceleration in application activity that continues until 2022, including the Covid-19 pandemic period. Notably, a substantial, discrete jump in applications is observed in 2023. These time-series patterns are shared by all AI business applications, those based on main business line, as well as high-propensity AI business applications.

Figure 1 also contains the annual number of employer startups that result from AI business applications within 8-quarters from the time of application. The time series contains actual startups for 2005-2019, and the projected startups for 2020-2023, for which actual startups that form within 8-quarters are not yet available. The projections are based on a linear probability model used by BFS that relates the event of startup formation to application characteristics.[26] By 2019, there were a total of about 1,300 startups that formed from AI applications (within 8 quarters of application) over the period since 2005, with much more (nearly 1,700) projected for the 2020-2023 period – consistent with the surge in applications in recent years. In general, all series follow similar time-paths, as shown in Figure A.1 in Appendix A, where each series is indexed to 100 for the year 2005. For comparison, A.1 also includes the time-series for all business applications in the U.S. While the number of all applications nearly doubles during the pandemic and stays elevated, this growth pales in comparison to the growth in AI business applications.

The patterns in Figures 1 and A.1 generally conform with typical early phases of industry life-cycles during which entry escalates, and the early diffusion of innovations during which adoption rises – as documented in Gort and Klepper (1982), Rogers (1962), and the related literature.[27] We summarize our main finding in this section below.

Finding 1 *AI business applications and formations have been increasing rapidly in recent years, with a notable jump in 2023.*

[26] See Bayard et al. (2018); Dinlersoz et al. (2021) for the details on the models used to estimate the likelihood of transition to an employer business within 8 quarters of the application.

[27] See, e.g., Jovanovic and MacDonald (1994), Klepper and Simons (2000) and Dinlersoz and MacDonald (2009).

4.2 Characteristics of AI business applications

A key question is whether AI business applications systematically differ from other applications. In particular, do AI business applications *disproportionately* possess early-stage characteristics associated with a higher potential to become employer businesses? Do they differ from other applications in terms of their expected startup employment? In other words, do they represent more promising entrepreneurship in terms of job creation?

Table 2 indicates that around 42.9% of all non-AI business applications are high-propensity applications, which have on average a 23.9% share of transitioning into employer businesses as opposed to about 5.3% for non-high propensity applications. For AI business applications, the fraction that are high-propensity is 50.7% – about 18% (or 7.8pp) higher compared to non-AI applications. This fraction differs only slightly across the two AI application types: 51.3% for those identified based on main business line, and 50.8% for those identified based on business/trade name.

There is less of a separation in terms of indicating a first planned wage payment date (wage). 18.5% of non-AI business applications indicate planned wages, while 19.1% of AI business applications fall into this category – with modestly higher rates for applications based on main business line (20.3%) compared to 17.8% for business/trade name only. We note that applications with a planned wage date overall have a relatively high (37.4%) likelihood of transitioning into employer status, versus 7.8% for applications without planned wages.

Notably, AI business applications are disproportionately composed of applications made by corporate entities (corp). 19.2% of non-AI business applications are from corporations, whereas nearly 40% of AI business applications are from corporations, with similar rates for those identified based on main business line versus business/trade name only. Applications from corporations have in general about 25.8% likelihood of transitioning into employer startups versus 10.3% for other legal forms of organization.

Next, we further unpack the connection between AI business applications and application characteristics using regression analysis. Table 3 contains the estimates from Linear Probability Models (LPMs). Unconditionally, a high-propensity application is associated with a 0.011 percentage points higher likelihood of being identified as an AI application (column 1) – equivalent to the 32% of the overall rate of AI application incidence (0.035%) among business applications. Conditional on detailed industry (4-digit NAICS), state, year, and application week (of the year) controls (column 2), a high-propensity application has 0.037pp higher likelihood of being an AI application (equivalent to 105% of the overall rate of AI application incidence).

Controlling further for more detailed application characteristics instead of the high-propensity indicator (column 3), we observe that many have a significant positive association with an AI application. These application characteristics are indication of planned wage payments (wage),

being a corporate entity (corp) – versus the omitted category of other legal organization forms, indicating starting a new business or hiring an employee as a reason for application (new bus./hire) – versus the omitted category of other reasons for applying (such as banking or compliance with IRS), and having an executive or trade name (exec./trade name). In terms of the magnitudes, applications made by corporate entities stand out. An application made by a corporation (as opposed to other legal forms of organization) has nearly 0.05pp higher likelihood of being an AI application – equivalent to 140% of the overall rate of AI application incidence.

Doing the same analysis for AI business applications based on main business line versus business/trade name only results in qualitatively similar findings, but different magnitudes (columns 4-9). Our main finding is summarized below.

Finding 2 *AI business applications disproportionately possess characteristics associated with higher likelihood of becoming employer businesses.*

As an intensive measure of a business application's prospects of job creation, we next consider the expected initial size of the planned business, measured by the highest (maximum) number of employees expected in the first 12 months following the application, as reported by the applicants on the application form. This variable is only available from 2017 onwards in our sample. It takes on a value of zero if the applicant does not expect to hire in the first 12 months – e.g., if the applicant is aiming for a nonemployer business or does not plan to hire in the initial stages of an intended employer business. Otherwise, it is a positive integer that provides the applicant's subjective estimate of the maximum initial size of the intended business.[28] As shown in Table 2, the average expected size for non-AI business applications is about 0.62 employees – a figure dominated by the large presence of likely non-employer business applications. AI business applications, however, expect about 1 employee on average in their first year - nearly 60% higher than the average for non-AI applications. The expectations are even higher (1.15 employees) for AI business applications based on main business line, but lower (0.82 employees) for those based on business/trade name only.

Table 4 considers the relationship between application characteristics and the expected initial size using an OLS regression. To accommodate the cases with zeros (i.e. no employment expected), the dependent variable is the "TVV" or "DHS" transformation (instead of a log transformation) of the expected maximum number of employees.[29] Controlling for detailed industry, state, application year, and application week (column 1), an AI application is associated with 24% higher expected

[28] There is a small number of outliers in this variable that indicate initial expected maximum employment values over 500. To ensure that these outliers do not influence our analysis unduly, we winsorize this variable to 100 employees. The results are very similar when a lower threshold of 50 employees is used.

[29] Specifically, the transformation is $\widetilde{y} = 2\frac{(y-\bar{y})}{(y+\bar{y})}$, where $\bar{y}$ is the grand mean of the expected initial size in the entire sample—for further information on this transformation, see Tornqvist et al. (1985) and Davis et al. (1996).

initial size.[30] The expectations are 29% higher for AI applications based on main business line, and 19% higher for AI applications based on business/trade name only (column 2). Once we control for application characteristics (columns 3 and 4), AI business applications have only 3.5% higher expected initial scale (about 4% higher for main business line cases and 3% for business/trade name cases). Including other application characteristics substantially attenuates the estimated coefficients for AI application indicators. In other words, the characteristics of the AI business applications that are associated with higher transition rates documented earlier accounts for most of the expected initial size premium associated with an AI application.[31] We summarize our main finding below.

Finding 3 *AI business applicants anticipate higher initial scale (employment) than non-AI business applicants.*

The analysis so far indicates that AI business applications disproportionately have characteristics that are positively associated with a higher potential for job creation. These findings are consistent with survey-based evidence in McElheran et al. (2024) that AI-using young firms (0-5 year old employer startups) have early characteristics associated with high-growth entrepreneurship. Next, we examine the transition rates of AI business applications to actual employer businesses.

4.3 Transition rates and average duration to transition

As shown in Table 2, the transition rate (with 8 quarters of application) of non-AI business applications to employer startups over the period 2004q3-2019 is 13.3%. AI business applications have 1.5pp (or 11.2%) higher transition rate at 14.8%. The transition rates for AI business applications based on main business line versus business/trade name are 14.0% and 15.8%, respectively. We also observe that during the period 2004q3-2019, the transition rate of AI business applications is stable, whereas the transition rate of other applications drift persistently lower to about 8%, after a large drop at the beginning of the Great Recession. In other words, the gap between the transition rates of AI versus non-AI business applications is increasing. Overall, the higher transition rate of AI business applications is consistent with generally better transition potential of these applications based on their characteristics, as explored in the previous section.

Table 5 contains the estimates from LPMs that examine the association between the transition rate of AI business applications and application characteristics. AI business applications have

[30] The coefficient estimates are converted to percentages by reverse-transforming the DHS transformation into a percentage difference. For each coefficient, this percentage difference is $100 \times 2\beta/(2-\beta)$.

[31] Note, in particular, applications with planned wages (wage) tend to be associated with a much larger expected initial size. Among applications, indicating planned wages is highly correlated with stating a positive expected initial size, and including this indicator (and other application characteristics) explains a substantial fraction of the variation in expected initial size.

about 2.9pp higher transition rate, controlling for state, detailed industry, state, application year, and application week (column 1). This amounts to a 22% higher rate compared to the transition rate of all applications (13.3%). The transition rate is higher by 2.5pp (or 19% relative to the mean) for AI business applications based on main business line, versus by 3.4pp (or 26% relative to the mean) for those based on business/trade name only (column 2). However, once other observable application characteristics are controlled for, AI business applications appear to be no different statistically in terms of transition rate from the rest of the applications (column 3). In fact, AI business applications based on main business line have a slightly lower (by 1.2pp, or 9.0% relative to the mean) transition rate (column 4). In other words, the higher transition rate of AI business applications appear to be accounted for mainly by their observable characteristics that are correlated with a higher likelihood of transitioning – not just because they are designated as AI business applications.

Finding 4 *AI business applications are more likely to transition into employer businesses than non-AI business applications, mainly explained by their observable characteristics.*

There are several frictions associated with starting a business. Some of these frictions may be reflected in the duration of the pre-transition gestational state that a potential entrant is in after making a business application. The intended business may materialize shortly after the application or may take longer depending on the severity of the frictions as well as the nature and complexity of the business idea. Table 2 indicates that the average duration from application to transition, conditional on becoming an employer business within 8 quarters of application, is 1.77 quarters for non-AI applications and 2.15 quarters for AI applications. The average duration is slightly higher (2.18 quarters) for AI business applications based on main business line, and slightly lower (2.13 quarters) for those based on business/trade name only. In other words, AI business applications take slightly longer on average to transition into an employer business.

Table 6 relates the DHS-transformed duration measured in quarters to observable characteristics of an application. Once we control for industry, state, application year and week, and other application characteristics, AI business applications do not appear to have a statistically different duration. Note also that applications indicating a planned wage date generally take less time to transition, whereas applications from corporations and those aiming to hire or start a new business take longer to transition, all else constant. Overall, while AI business ideas take slightly longer to implement, they are not particularly tardy once their initial characteristics are controlled for.

Finding 5 *Conditional on becoming an employer business, AI business applications take longer to transition than non-AI business applications, mainly explained by their observable characteristics.*

4.4 The distribution of AI business applications across states

We now examine the distribution of AI business application activity across states. Figure 2 maps the percent of AI business applications originating from each state for the entire analysis period. The states with the largest shares are California (19.8%), Florida (9.4%), New York (8.3%), and Texas (7.5%). The distributions across states are similar for AI business applications based on main line of business versus business/trade name – see Appendix Figure A.2. To see the concentration of AI business applications relative to state size, we next normalize AI business applications by state population.

Figure 3 shows the number of AI business applications per capita (per 100,000 residents of age 18-64) based on the average population of each state over the period 2005-2022.[32] The states with the highest numbers of applications per capita are, by far, Delaware and Wyoming. Delaware stands out because AI business applications are disproportionately composed of applications from corporations, and Delaware receives a disproportionately high number of corporation applications owing to the favorable state corporate tax laws. Wyoming ranks high among states in terms of growth in overall business applications over the last two decades – it experienced a 12-fold increase in business applications since 2005.[33] Wyoming was also recently selected as a federal tech hub under the CHIP Act and has added new tech incubators (such as IMPACT 307), and it has attracted AI-related entrepreneurship in recent years.[34] These states are followed by District of Columbia, Nevada, California, Florida, and Utah. The rest of the states have much smaller AI business applications per capita. The patterns are similar for AI business applications based on main line of business versus business/trade name – see Appendix Figure A.3.

Finding 6 *AI business applications are concentrated across states, with California having the highest share of applications.*

4.5 The distribution of AI business applications across sectors

The share of AI business applications by sector are shown in Figure 4. The sectoral distribution is far more concentrated than the distribution across states explored in the previous section. In fact, a single sector stands out. The majority of applications (56%) of the applications is in Professional, Scientific, and Technical Services (NAICS 54). This is followed by Finance and Insurance (NAICS 52) and Information (NAICS 51), which contain 9.4% and 9.1% of the applications, respectively.[35]

[32] State-level population estimates by age were obtained from the Census Bureau.

[33] See Business Formation Statistics.

[34] See, e.g., 31 regional tech hubs and U of Wyoming tech hub.

[35] The high concentration in NAICS 54 is also part of the reason why AI business applications tend to be disproportionately high propensity business applications (HBA), since NAICS 54 contains two industries (NAICS 5411 - Legal Services and NAICS 5413 - Architectural, Engineering, and Related Services) that are classified as part of high-propensity applications.

Other notable sectors include Administrative and Support (NAICS 56), Retail (NAICS 44-45), Health Care (NAICS 62), Manufacturing (NAICS 31-33) and Other Services (NAICS 81) – each of these contain around 2-3% of total AI business applications. Overall, NAICS 51 through 56 account for almost 80% of AI business applications – 82.3% of applications based on main business line and 74.9% of applications based on business/trade name. These patterns may change, however, as diverse applications of AI technologies continue to emerge in other sectors (such as Arts and Entertainment, NAICS 71). These patterns are broadly similar to the sectoral distribution of AI-using firms observed in the 2018 and 2019 ABS, and more recently in the BTOS for 2023 and 2024 – see Acemoglu et al. (2022), McElheran et al. (2024), and Bonney et al. (2024). However, a key difference is the much higher concentration in NAICS 54.

Finding 7 *AI business applications across sectors are highly concentrated, particularly in Professional, Scientific, and Technical Services (NAICS 54).*

4.6 External validation and supporting evidence

Because our data and measurement are novel, we now perform a number of exercises to provide additional support for the evolution of AI business application and startup activity we documented so far. These exercises indicate that the patterns we observe are consistent with the development and commercialization of AI technologies over the last two decades, and the independent evidence from AI-related patent and trademark activity and firm-level AI adoption patterns in recent Census Bureau surveys.

4.6.1 Timing with respect to the milestones in recent AI history

The patterns of growth in AI business activity and the turning points in the time series are consistent with the timing of key developments in AI technologies in the last two decades. The pre-growth period (2005-2011) in AI business application activity saw a few key early developments that set the stage for major breakthroughs later in the 2010s and 2020s. For example, in 2005, Stanford University's autonomous vehicle, Stanley, demonstrated the potential of AI in autonomous driving. In 2006, IBM Watson completed initial tests for competing in Jeopardy, eventually winning in 2011. The early methods to scale unsupervised learning to big data were also developed during this period.[36] There were also notable advances in 2011 in speech recognition and computer vision – e.g., the release of Siri by Apple and Kinect for Xbox 360 by Microsoft.[37]

[36] For instance, Raina et al. (2009) presented the idea of using Graphics Processing Units (GPUs) to train large neural networks.

[37] The latter featured a machine learning application for human motion capture via computer vision.

2012 was a key year in AI history. Krizhevsky et al. (2012) introduced a deep convolutional neural network that won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) for image classification and led to the explosion of deep learning research. Over the next few years, other deep neural networks were developed.[38] These networks offered outstanding performance for traditional machine classification algorithms, enabling further applications of deep learning. Perhaps not coincidentally, the first turning point in the AI business applications series occurs in 2012: the annual flow of new business applications starts to increase, albeit slowly.

Developments in AI continued in 2013, as Word2vec by Google was introduced to automatically identify semantic relationships between words, a key development in natural language processing.[39] In 2014, Goodfellow et al. (2014) invented generative adversarial networks, a class of machine learning approaches used to generate photos and transform images. Kingma et al. (2014) used variational methods to make generative approaches to supervised learning more efficient in large scale applications. Facebook (now Meta) developed the deep learning facial recognition system, DeepFace, to identify human faces in digital images.

2016 saw another major breakthrough with key developments in unsupervised and reinforcement learning. Notably, AlphaGo was able to play the game of Go at the level of top players, and make moves that these players could not even consider. Further advances in the ability of AI to imitate human behavior also took place in 2016. AI algorithms were developed to create text that mimics handwriting, and to imitate human speech (e.g., Google's Wavenet). Another significant development was Uber's initiation of a self-driving car pilot program in Pittsburgh. These advances synchronize with the acceleration in AI business application activity around 2016. Business applications started to arrive at a higher rate compared to the period 2012-2015. The applications of deep, unsupervised, and reinforcement learning methods to business problems propelled some of this growth.[40]

The capabilities of AI continued to improve after 2016. In 2017, researchers at Google introduced the concept of "transformers", inspiring the development of tools that could automatically parse unlabeled text into large language models (LLMs).[41] Advances enabled by transformer networks led to the release of the first generative pre-trained transformer (GPT-1) by OpenAI in 2018, followed in 2019 by GPT-2, which demonstrated the ability to generalize unsupervised to many different tasks. Meanwhile, on the medical front Google AI and Langone Medical Center's deep learning algorithm outperformed radiologists in detecting potential lung cancers.

[38] These include the Visual Geometry Group Very Deep Convolutional Network (VGGNet) and GoogLeNet.

[39] See, e.g., Mikolov et al. (2013).

[40] These applications include, but are not limited to, customer service and experience, quality control, manufacturing safety, predictive maintenance, fraud detection, stock trading and investment, pricing strategies, marketing automation, and medical diagnosis.

[41] See Vaswani et al. (2017).

The period 2020-2023 witnessed major advances in generative AI. In 2020, OpenAI released GPT-3, which is capable of generating human-like text. In 2021, OpenAI introduced Dall-E, an AI system that can generate images from text prompts. DallE contributed to the emergence of high-quality "AI art" from natural language prompts. Open AI's ChatGPT was released in November 2022 to provide a chat-based interface to its GPT-3.5. In 2023, OpenAI announced GPT-4, which receives both text and image prompts. Google also released its own chatbot (Bard), and Microsoft launched Bing Chat (later renamed Microsoft Copilot), a GPT-4-based chatbot integrated into the search engine.

The advances in generative AI in the early 2020s have broadened the applications of AI in business.[42] Perhaps not coincidentally, there is a discrete and substantial jump in AI business applications in 2023. The acceleration in AI business initiation activity during 2020-2023 was likely in part also driven by the Covid pandemic that increased incentives for automation more broadly. The rapid rise in AI business applications in recent years is also consistent with rising AI-related hiring, publications, conferences, courses, and other educational programs.[43] It is also consistent with an independent estimate of AI startups that received funding, which show an increase in recent years, and particularly in 2023 – similar to the jump we find for that year.[44] Overall, the patterns of growth in AI business applications appear to align with the timing of the development of AI technologies with broader business applicability.

4.6.2 Timing with respect to developments in complementary technologies

The time-path of AI business applications is also related to the development of complementary technologies enabling AI. Advances in digitization, big data, cloud storage and computing, and computing power during the last two decades have boosted AI development. These advances include the introduction of Simple Storage Service (S3) and Elastic Compute Cloud (EC2) in 2006 by Amazon's AWS, followed by the launch of Google App Engine in 2008, Microsoft Azure in 2010, and Microsoft Office SaaS on 2011. AI capabilities have been increasingly integrated with cloud services, enabling new possibilities for automation, analytics, and decision making. Similarly, increases in computing power and speed have enabled the implementation of large scale AI systems with many commercial applications.[45]

Consistent with the data requirements and computing demands of AI, recent analysis using

[42] Some of these applications are more sophisticated customer service, marketing automation, translation services and multilingual customer support, content creation, sentiment analysis and audience research, survey design, document review and analysis, compliance assistance, automated reporting, competitor intelligence, and the design of product catalogs and descriptions.

[43] See Babina et al. (2020, 2022) and Artificial Intelligence Index Report 2023.

[44] See Artificial Intelligence Index Report 2024.

[45] More powerful microchips to meet the processing demands of AI have been introduced recently, with the primary example being Nvidia's A100 GPU released in 2020.

Census Bureau surveys (McElheran et al. (2024) and Zolas et al. (2020)) also shows that AI-using firms tend to have higher reliance on digitization and cloud services. The advances in these complementary technologies have undoubtedly contributed to the boom in AI business initiations.

4.6.3 Timing with respect to AI-related patents and trademarks

Additional validation of the time series for AI business applications comes from the data on intellectual property (IP) related to AI. Innovations in AI technologies by both existing and new businesses as reflected in AI-related patents can be a precursor of AI business applications. Furthermore, many of the emerging AI businesses need to protect their innovations, products, services, processes, and brand names by obtaining patents and trademarks early in their life-cycles. An escalation in the IP activity is thus expected to accompany the rise in AI business applications.

As a measure of innovations in AI, we use the AI patents identified by Branstetter et al. (2023) with a high confidence score (95%). AI-related trademark applications and registrations were identified by searching the trademark textual information in USPTO's Trademark Case File Database (TCFD) for AI-related keywords based on the same dictionary that was used to identify AI business applications – hence similar caveats apply as in our identification of AI applications. Note also that AI patents and trademarks include IP activity by existing firms, not just by new businesses.

The time-patterns in AI business application activity are overall consistent with those in AI-related patent filings and trademark activity shown in Figure 5. Over the period 2005-2018, the contemporaneous correlation between annual AI applications and AI patents granted is 0.78. The correlation between annual AI applications and AI trademark applications (registrations) is 0.98 (0.97). Figure A.4 shows a normalized version, where each series is indexed to 100 for 2005 for easier comparison of trends. AI business application activity appears to lag AI-related patenting considerably. AI business applications seem to have a shorter lag with AI-related trademark applications and trademark registrations. Trademark applications and registrations start to take off around 2010, about two years before AI business applications start to rise. The contemporaneous correlation of the annual growth rate of AI business applications with that of AI trademark applications (registrations) is 0.49 (0.52) over the period 2006-2018. In general, trademark applications are an early indicator (see deGrazia et al. (2020)), and registrations may have a shorter lead over business formation activity since a registration requires proof of use of the trademark in commerce. Overall, the patterns in AI-related patenting and trademarking activity generally lend further support to the time-path of the AI business application activity.

4.6.4 AI adoption patterns in recent Census Bureau surveys

Recent evidence from Census Bureau surveys measuring AI use and adoption indicates a rise in the adoption rate. The use rate for the period 2016-2018 is estimated at 3.2% based on the 2019 ABS.[46] Most recent high-frequency measures of AI use rates in BTOS, on the other hand, indicate estimates close to 5.5% as of late 2023 – a 72% increase over the last 5-years.[47]

The increase in AI business applications and formations is consistent with the growth in adoption rate in recent years. Business applications are an early indicator of new business formation and non-farm employment growth.[48] The continuing rise in AI business applications therefore suggests an increasing contribution to the adoption rate of AI by new AI businesses in the near term. Consistently, AI use rate among the smallest firms (1-4 employees) in BTOS also appears to be accelerating: about 6.0% as of late 2023, and expected to increase to about 8.1% by the end of 2024.

The sectoral and geographic distribution of AI business application activity also largely conforms to the patterns in recent Census Bureau surveys. Based on the analysis of the 2019 ABS, AI use is most common in Information (NAICS 51), Professional, Scientific, and Technical Services (NAICS 54), and Finance and Insurance (NAICS 52).[49] More recent estimates of the sectoral AI use rates from BTOS indicate that NAICS sectors 51-56 exhibit high rates; Information (NAICS 51) leads, followed by Professional, Scientific, and Technical Services (NAICS 54). AI business application activity is also concentrated in these sectors, albeit with a much heavier presence in sector 54 compared to the ABS and BTOS – most business applications developing or providing AI technologies likely fall into these sectors. Similarly, many states with high AI business applications per capita, such as California, Florida, Nevada, Utah, are also among the ones with relatively high rates of AI use among firms based on the BTOS.

4.7 Performance of AI businesses

We now turn to the outcomes of employer businesses (startups) that originate from AI business applications at any point in time during the period 2004q3-2021. We compare the performance of these businesses over their life-cycles with that of the businesses originating from non-AI business applications. We emphasize that our analysis does not aim to identify causal effects of AI presence on business outcomes. Instead, it seeks to explore the distinguishing characteristics of businesses

[46] See Acemoglu et al. (2022). Zolas et al. (2020) also provides estimates of adoption rates in 2017 for some AI-related technologies based on the 2018 ABS.

[47] Estimates we reported based on BTOS public data exclude "don't know" answers to current and future AI adoption questions to be consistent with the estimates generated earlier based on the 2019 ABS, which exclude "don't know" answers.

[48] See Asturias et al. (2021, 2023).

[49] See Acemoglu et al. (2022).

that emerge from AI-related business applications.

For post-transition outcomes, we follow an application after it is first observed as an employer business in the LBD until it exits (i.e. it ceases to have positive employment based on the Business Dynamics Statistics' (BDS) definition of exit) or is involved in a merger and acquisition (M&A). There is no formal definition of M&A event in the LBD and such events are hard to identify precisely based on the available administrative data. To analyze a firm's outcomes before it is absorbed into another business as a result of M&A, we define what we call "organic firm growth" before exit. Organic growth includes the events of (i) continuing as a single unit firm under the same firm identifier, (ii) transitioning with a different firm identifier to another single unit firm or to a multi-unit firm based on the following criteria: the multi-unit firm has three or fewer establishments at the time of transition, the multi-unit firm has its first employment in the LBD at most 2 years prior to the year of transition, and only a single EIN is associated with the multi-unit firm. We refer to event (ii) above as "firm transformation". In other words, our definition of firm transformation is one where a firm grows or undergoes changes without a break in its activity due to being absorbed in another business entity. The rest of the events a single-unit firm experiences are considered as "M&A" activity.

We found that M&A activity for AI businesses over this time period is too rare to inform general trends on the types of business application characteristics that lead to this type of activity. If and when M&A activity for AI businesses picks up in the coming years, we will be able to explore this type of event further. We note that our findings are largely robust to some alternative ways of defining organic firm growth – for instance excluding transitions to another single unit firm. This is largely due to the fact that firm transformation instances in our sample are relatively rare, and do not substantially influence the general patterns we discover.

Many AI businesses form after 2016 when AI business applications start to arrive at a higher rate. Since our outcome measures are available until 2021 (the last year for which the LBD is available), we are effectively documenting the early life-cycles (up to 5 years after birth) for many of the AI businesses – we have a longer horizon for AI businesses that formed earlier in the sample period.

Table 7 contains the estimates from OLS regressions that relate several key firm outcomes to application characteristics in a panel that tracks startups over time after formation. All continuous dependent variables are DHS-transformed.[50] Two dependent variables are indicators (analyzed using LPMs). One is the indicator of firm exit, and the other is the indicator of firm transformation as described above. We also used an M&A indicator separately as a dependent variable, but found that there is a very small number of M&As in the case of businesses originating from AI-business

[50] All continuous dependent variables in Table 7 were winsorized at 99.9% to reduce the influence of some gross outliers.

applications – potentially because we are still in the early phases of the diffusion of AI.[51] In all specifications in Table 7, we control for initial application characteristics and a large number of fixed effects, including industry-year fixed effects to account for time-varying factors at the industry level – importantly, the latter also allows us to control for industry specific time-varying price levels in our analysis of (nominal) firm revenue.

First, consider the firm size measures in columns 1-3. AI businesses tend to exhibit 15% higher employment, 39% higher payroll, and 13% higher revenue.[52] Note also that initial application characteristics have significant positive relationships with firm size measures. For instance, indicating a planned wage payment date is associated with a 17% higher employment, and applications from corporations are associated with nearly 3% higher employment. Similarly, when the reason for application is to hire or start a new business, employment is higher by about 5%. Having a previous EIN and an executive or trade name are associated with 8% and 13% higher employment, respectively – these variables may be correlated with prior business experience and more concrete business plans, respectively. Overall, initial characteristics tend to be significantly related to the scale of businesses originating from the applications.[53] The better performance of AI businesses is also consistent with recent survey findings that AI use is associated with better overall business performance and increases in employment and revenue – see Bonney et al. (2024) for evidence from the Census Bureau's Business Trends and Outlook Survey (BTOS).

Because we mainly document the early-life cycles for most of the AI businesses, they may not exhibit exceptionally high profitability and productivity during these early years. This would be the case if they are primarily engaged in developing new products and processes integrating AI technologies without necessarily generating high revenue initially relative to their input costs. Table 7 relates the annual measures of average payroll (payroll/employment), labor share (payroll/revenue) and labor productivity (revenue/employment) outcomes to initial application characteristics (columns 4-6). Conditional on all other controls, AI businesses have 29% higher average payroll and 27% higher labor share compared to other businesses. The larger average pay and labor share we observe is likely attributable to the generally higher skill level of AI businesses' workforce. Consistently, Acemoglu et al. (2022) find that firms indicate an increase in the overall skill of their employees due to AI use. However, we do not observe a significantly higher labor productivity for AI businesses at this early stage of AI diffusion – consistent with their higher employment but not as much higher revenue (columns 1 and 3). This may not be too surprising as startups early in their life cycle may be more grow-oriented, rather than targeting productivity increases.

[51] Disclosure restrictions by Census Bureau prevent us from presenting statistics on the M&A activity we identified.

[52] These estimates are obtained by converting the estimated coefficients from DHS transformation to percent differences.

[53] See also Table 9 in Bayard et al. (2018) for descriptive relationships between application characteristics and initial employment of startups.

Column (7) in Table 7 examines the relationship between business exit (failure) in any given year and application characteristics. Conditional on controls, AI businesses exhibit 3pp higher exit rate, or about 25% higher compared to the average annual exit rate of about 12% in the overall sample. Finally, in Column (8) we do not find any statistically significant difference between the transformation rates for businesses originating from AI business applications versus others. We summarize below our key findings in this section.

Finding 8 *Controlling for application characteristics, AI businesses have higher employment, revenue, labor share, and average pay, but not higher labor productivity compared to non-AI businesses. They also exhibit higher exit rates.*

4.8 Estimates of AI Use Rate in the Near Term

We next use the law of motion (4) to provide some back of the envelope estimates of AI usage rates among all businesses for the period 2023-2027. We estimate various components of (4) using our current findings, while incorporating data from the Business Trends and Outlook Survey (BTOS) and Business Dynamics Statistics (BDS) – see Table 8.

First, consider the numerator of (4). As a starting point, BTOS provides estimates for AI adoption rate at the end of 2023: $a_{2023} = 5.5\%$. Based on BTOS, Bonney et al. (2024) estimates the de-adoption rate in 2024 ($d_{2024} = 14.5\%$), and the adoption rate in 2024 by incumbent businesses that have not yet adopted ($r_{2024} = 3.3\%$). We next apply the exit rate of 12% for AI businesses estimated from Table 7 using recent years.[54] This rate implies $\chi^a_{2024} = (X^a_{2024}/N^a_{2024}) \times a_{2024} = 0.12 \times 0.055 = 0.066$. While these rates are likely to change over time, we do not yet have reliable estimates for the near term. Therefore, we assume that the rates d, r, and χ^a remain the same at their 2024 values over the period 2025-2027.

For entry, we start with the number of AI startups we estimate for 2024 based on the estimated average annual growth rate of AI startups over the period 2012-2023 (31% from Figure 1). Based on the most recent trends in Figure 1, this growth rate may be low but assuming it persists, we estimate the number of startups and the resulting entry rates for the years 2025-2027. We also use two alternative growth rates in the number of AI startups – one based on the most recent growth rate in 2023 (100%), and another one that is optimistic (200%). These high rates may materialize, at least in the near term, depending on whether the high growth rate for 2023 (106%) persists or increases. Finally, to estimate the growth rate of the total number of firms in the economy (g_{t+1}) in the denominator of (4), we use the average annual growth rate of the number of firms from the

[54] This is based on the overall firm exit rate for recent years (about 9%) and the estimated 3 percentage points higher exit rate for AI businesses.

Business Dynamics Statistics (BDS) over the 5-year period 2017-2021 (0.56%) – 2021 is the latest year available in the BDS.

Using these components of the law of motion (4), we simulate the adoption rate starting at the end of 2023. The results are shown in Figure 6. In all three growth rate scenarios, despite the large differences in the growth rates of AI startups, the overall adoption rate reaches 9-10% by the end of 2027. This represents almost a doubling of the rate in 2023, but not a very high rate of adoption in the near term. We emphasize that this conclusion can change depending on how various components of the law of motion deviate from our baseline assumptions. As demonstrated in this section, we can use our conceptual framework to track the diffusion of AI as more frequent and reliable measures of adoption rates and de-adoption rates by existing firms, as well as entry, and exit rates become available.

5 Summary and Discussion

This paper has documented some key facts about early-stage AI-related business activity. The findings point to a surge in AI business applications in the last decade, with particularly rapid growth in recent years. AI business applications are, on average, more likely to transition to employer businesses, and exhibit better post-transition outcomes in some dimensions compared to other employer businesses – e.g. employment, pay, and revenue. AI businesses originating from applications appear to form a dynamic segment of employer business population in terms of turnover (entry and exit). This dynamic view is further supported by recent research that finds that AI-using startups tend to exhibit faster growth early on in their life-cycles compared to other startups – see McElheran et al. (2024).

While emerging AI businesses do not exhibit higher labor productivity compared to other businesses that form during the same period, this pattern may change as more AI businesses mature. The higher labor share and average pay of these businesses suggest an early stage during which many AI startups focus on developing processes or products by relying on high-skill labor that leads to a higher labor cost relative to revenue, but not higher labor productivity.[55] Labor productivity may rise over time relative to other businesses, as some surviving AI startups grow and generate increasing revenue relative to their labor input. In support of this possibility, Acemoglu et al. (2022) find, based on the 2019 ABS, that labor productivity increases with firm size, and AI-using firms as a whole (including larger and older firms that use AI, not just young or small firms) have about 2% higher labor productivity conditional on firm characteristics – though this premium is not precisely estimated.

[55]Note that, by definition, a firm's labor productivity (revenue/employment) is the product of the inverse of its labor share (payroll/revenue) and its average payroll (payroll/employment).

Overall, the emerging landscape for AI business applications and startups suggests a small but growing contribution of AI business application activity to job-creating entrepreneurship and business dynamism. This optimistic picture, however, has to be interpreted with caution from the perspective of industry life-cycles and the diffusion of innovations. We are still at an early stage of the emergence of an "AI business ecosystem" in the U.S. – including providers and users of AI. The broad literature on industry life-cycles suggests that the initial accelerated entry may slow down. A shakeout may follow at least for some types of AI-related businesses. For instance, rapidly increasing demand for data and computation power for AI training can become, if not met with an adequately growing supply, a constraint that impedes further entry and triggers a shakeout – see, e.g., Bessen et al. (2022), Lohn and Musser (2022), and Lohn (2023). Technological constraints, data requirements, and emerging regulatory environment each influence the diffusion of AI, which is ultimately a function of the distinct benefits and costs of AI for businesses, consumers, and workers, and how these benefits and costs change over time – see, e.g., Agrawal et al. (2023a,b); Svanberg et al. (2024).

6 Conclusion

Despite the growing interest in the rise of AI and its effects on businesses and workers, information on its diffusion among businesses has been limited. We provide new evidence on this front by using comprehensive administrative data on business applications to identify emerging AI business ideas and startups originating from these ideas over the period 2004-2023. Our methodology relies on textual information revealed by applicants about their prospective products and services, as well as the names of their businesses. While our approach does not necessarily pin down the *number* of all business applications aiming to develop or use AI, the *growth* patterns we observe are informative about the acceleration in new business activity related to AI, as several validation exercises suggest.

We document a sharp rise in AI business applications in recent years that may persist into the near future as new AI technologies with applications to business processes continue to emerge. We find that AI business applications are associated with high potential entrepreneurship. Businesses originating from these applications generally perform better in some key business outcomes.

The analysis demonstrates the usefulness of BFS micro data in exploring the emergence of new technologies and the evolution of new industries. An initial phase of entry is an important stage of this evolution. BFS micro data can be used for early detection and analysis of this phase.

Our analysis can also be used to generate high-frequency and timely information on the diffusion of AI among early-stage businesses. For example, a monthly series of AI business applications can be constructed. Our approach to identifying AI business applications can be updated over time to keep

this series current and relevant as new AI technologies/applications appear. Furthermore, similar approaches can be applied to other emerging technologies, opening the possibility of building a high-frequency, near real-time index of the diffusion of new technologies among early-stage businesses.

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Table 1: AI Business Applications

Type of Identification	# of AI Applications	% of All AI Applications	% of All Applications
Unfiltered (Raw) AI applications			
AI - Main Business Line	11,970	43.7%	0.019%
AI - Business Name	15,872	57.9%	0.026%
AI - Trade Name	1,699	6.2%	0.003%
AI - All	27,370	100%	0.044%
Applications filtered out	5,639	20.6%	0.009%
Filtered (Core) AI applications			
AI - Main Business Line	11,545	53.1%	0.019%
AI - Business/Trade Name	10,186	46.9%	0.016%
AI - All	21,731	100%	0.035%

Notes: Includes all business applications for the period 2004q3-2023m10. For unfiltered (raw) AI business applications, percentages of all AI business applications do not add up to 100% because some applications can be in two or more identification types (e.g., main business line and trade name). For filtered (core) AI business applications, applications based on business or trade name (AI - Business/Trade Name) are the complement set of applications based on main business line (AI - Main Business Line).

Table 2: Characteristics of AI Business Applications

	Non-AI	AI		
		All	Main Bus. Line	Bus./Trade Name
% high propensity	42.9%	50.7%	51.3%	50.8%
% wage	18.5%	19.1%	20.3%	17.8%
% corp	19.2%	39.9%	39.4%	40.5%
avg. exp. max initial emp.	0.62	1.00	1.15	0.82
transition rate (8 qtr. window)	13.3%	14.8%	14.0%	15.8%
avg. duration (qtrs)	1.77	2.15	2.18	2.13

Notes: The analysis covers business applications for the period 2004q3-2023m10 and employer business formations for the period 2004q3-2019q4. "AI All", "AI Main Bus. Line", "AI Bus./Trade Name" indicate, respectively, an AI business application, an AI application based on main business line, and an AI application based on business/trade name. Non-AI indicates applications not classified as an AI business application. The application types are high-propensity business applications (high propensity), applications with planned wages (wage), and applications from corporations (corp). "Avg. exp. max initial emp." is the average expected maximum employment in the 12 months following the application. Transition rate is the fraction of applications that transition to an employer startup within a window of 8 quarters from the application. Avg. duration is the average duration measured in quarters (qtrs) from the quarter of application to the quarter of transition, conditional on transitioning within 8 quarters.

Table 3: AI business applications and application characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	AI	AI	AI	AI	AI	AI	AI	AI	AI
	All	All	All	Main Bus. Line	Main Bus. Line	Main Bus. Line	Bus./Trade Name	Bus./Trade Name	Bus./Trade Name
high propensity	0.0111***	0.0369***		0.0063***	0.0195***		0.0049***	0.0174***	
	(0.0005)	(0.0008)		(0.0004)	(0.0006)		(0.0003)	(0.0005)	
wage			0.0096***			0.0071***			0.0025***
			(0.0007)			(0.0005)			(0.0004)
corp			0.0486***			0.0246***			0.0240***
			(0.0009)			(0.0007)			(0.0006)
new bus./hire			0.0105***			0.0052***			0.0053***
			(0.0006)			(0.0005)			(0.0004)
prev. ein			-0.0012			-0.0016			0.0004
			(0.0014)			(0.0010)			(0.0009)
exec./trade name			0.0070***			0.0031***			0.0039***
			(0.0007)			(0.0005)			(0.0005)
bus. address			-0.0006			-0.0008			0.0002
			(0.0007)			(0.0005)			(0.0005)
year controls		*	*		*	*		*	*
state controls		*	*		*	*		*	*
week controls		*	*		*	*		*	*
industry controls		*	*		*	*		*	*
R^2	0.000009	0.00349	0.00352	0.000005	0.00220	0.00222	0.000004	0.001806	0.001822
N	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000	62,070,000

Notes: Includes all business applications for the period 2004q3-2023m10. Each column corresponds to estimates from a Linear Probability Model. The dependent variables are: AI All – indicator of an AI business application, AI Main Bus. Line – indicator of AI application based on main business line, AI Bus./Trade Name – indicator of AI business applications based on business/trade name. The coefficient estimates are expressed in percentages. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 4: Expected initial size and application characteristics

	(1)	(2)	(3)	(4)
	Employment	Employment	Employment	Employment
ai (all)	0.2161***		0.0346***	
	(0.0092)		(0.0017)	
ai (main bus. line)		0.2566***		0.0386***
		(0.0133)		(0.0024)
ai (bus./trade name)		0.1715***		0.0303***
		(0.0128)		(0.0023)
wage			3.0235***	3.0235***
			(0.0003)	(0.0003)
corp			0.0195***	0.0195***
			(0.0002)	(0.0002)
new bus./hire			-0.0180***	-0.0180***
			(0.0001)	(0.0001)
prev. ein			-0.0015	-0.0015
			(0.0028)	(0.0028)
exec./trade name			0.0291***	0.0291***
			(0.0002)	(0.0002)
bus. address			0.0150***	0.0150***
			(0.0001)	(0.0001)
year controls	*	*	*	*
state controls	*	*	*	*
week controls	*	*	*	*
industry controls	*	*	*	*
R^2	0.0750	0.0750	0.9632	0.9632
N	28,750,000	28,750,000	28,750,000	28,750,000

Notes: Includes all business applications for the period 2017m4-2023m10 – data for expected max initial employment starts in 2017m4. Each column corresponds to estimates from an OLS regression. The dependent variable is DHS-transformed expected maximum initial employment, winsorized at 100 employees at the top. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), an AI application based on main business line (ai (main bus. line)), and AI applications based on business/trade name (ai (bus./trade name)), high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 5: Transition rate and application characteristics

	(1)	(2)	(3)	(4)
	Employer Business	Employer Business	Employer Business	Employer Business
ai (all)	0.0290***		-0.0040	
	(0.0037)		(0.0035)	
ai (main bus. line)		0.0249***		-0.0128***
		(0.0049)		(0.0047)
ai (bus./trade name)		0.0342***		0.0069
		(0.0058)		(0.0054)
wage			0.2481***	0.2481***
			(0.0002)	(0.0002)
corp			0.0977***	0.0977***
			(0.0002)	(0.0002)
new bus./hire			0.0150***	0.0150***
			(0.0002)	(0.0002)
prev. ein			-0.0126***	-0.0126***
			(0.0005)	(0.0005)
exec./trade name			0.0009***	0.0009***
			(0.0001)	(0.0001)
bus. address			0.0096***	0.0096***
			(0.0002)	(0.0002)
year controls	*	*	*	*
state controls	*	*	*	*
week controls	*	*	*	*
industry controls	*	*	*	*
R^2	0.0888	0.0888	0.1930	0.1930
N	42,580,000	42,580,000	42,580,000	42,580,000

Notes: Includes all business applications for the period 2004q3-2019q4. Each column corresponds to estimates from a Linear Probability Model. The dependent variable is indicator of transitioning to an employer business within 8 quarters of the application. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), an AI application based on main business line (ai (main bus. line)), and AI applications based on business/trade name (ai (bus./trade name)), high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 6: Duration to transition and application characteristics

	(1)	(2)	(3)	(4)
	Duration to Transition	Duration to Transition	Duration to Transition	Duration to Transition
ai (all)	0.0290		0.0679	
	(0.0567)		(0.0548)	
ai (main bus. line)		0.0550		0.1055
		(0.0768)		(0.0751)
ai (bus./trade name)		-0.0002		0.0258
		(0.0838)		(0.0799)
wage			-1.0627***	-1.0627***
			(0.0017)	(0.0017)
corp			0.1883***	0.1883***
			(0.0020)	(0.0020)
new bus./hire			0.2378***	0.2378***
			(0.0024)	(0.0024)
prev. ein			-0.0037	-0.0037
			(0.0046)	(0.0046)
exec./trade name			0.1880***	0.1880***
			(0.0021)	(0.0021)
bus. address			-0.0219***	-0.0219***
			(0.0021)	(0.0021)
year controls	*	*	*	*
state controls	*	*	*	*
week controls	*	*	*	*
industry controls	*	*	*	*
R^2	0.0504	0.0504	0.1220	0.1220
N	5,666,000	5,666,000	5,666,000	5,666,000

Notes: Includes all business applications for the period 2004q3-2019q4. Each column corresponds to estimates from an OLS regression. The dependent variable is DHS-transformed duration (in quarters) from application to transition, conditional on transitioning within 8 quarters of application. Robust standard errors are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), an AI application based on main business line (ai (main bus. line)), and AI applications based on business/trade name (ai (bus./trade name)), high-propensity business applications (high propensity), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). Other controls are application year, state, week (of the year), and industry (4-digit NAICS). Observation counts (N) are rounded to protect confidentiality.

Table 7: Business outcomes and application characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Employment	Pay	Revenue	Pay/Emp	Revenue/Emp	Pay/Revenue	Exit	Transformation
ai (all)	0.136***	0.334***	0.124**	0.257***	-0.001	0.242***	0.030***	-0.001
	(0.0293)	(0.0351)	(0.0512)	(0.0205)	(0.0374)	(0.0329)	(0.0057)	(0.0011)
wage	0.160***	0.145***	0.083***	0.009***	-0.069***	0.074***	-0.004***	0.001***
	(0.0008)	(0.0009)	(0.0011)	(0.0006)	(0.0008)	(0.0007)	(0.0001)	(0.0000)
corp	0.030***	0.110***	-0.021***	0.132***	-0.013***	0.098***	-0.022***	-0.006***
	(0.0010)	(0.0012)	(0.0013)	(0.0007)	(0.0010)	(0.0009)	(0.0002)	(0.0001)
new bus./hire	0.047***	0.073***	0.061***	0.024***	-0.003**	0.045***	0.022***	0.0004***
	(0.0012)	(0.0013)	(0.0016)	(0.0008)	(0.0012)	(0.0010)	(0.0002)	(0.0001)
prev. ein	0.081***	0.064***	0.067***	-0.001	-0.005**	0.014***	0.002***	-0.0008***
	(0.0025)	(0.0028)	(0.0030)	(0.0014)	(0.0022)	(0.0018)	(0.0003)	(0.0001)
exec./trade name	0.126***	0.146***	0.086***	0.061***	0.008***	0.031***	-0.001***	0.001***
	(0.0011)	(0.0012)	(0.0014)	(0.0008)	(0.0011)	(0.0009)	(0.0002)	(0.0000)
bus. address	0.011***	-0.016***	-0.002	-0.036***	-0.018***	-0.016***	0.002***	0.001***
	(0.0010)	(0.0012)	(0.0014)	(0.0007)	(0.0011)	(0.0009)	(0.0002)	(0.0001)
BFS year FE	*	*	*	*	*	*	*	*
industry x LBD year FE	*	*	*	*	*	*	*	*
week FE	*	*	*	*	*	*	*	*
state FE	*	*	*	*	*	*	*	*
age categories	*	*	*	*	*	*	*	*
firm size categories				*	*	*	*	*
R^2	0.183	0.147	0.181	0.262	0.258	0.222	0.032	0.005
N	30,900,000	30,900,000	15,800,000	30,900,000	15,800,000	15,800,000	30,900,000	30,900,000

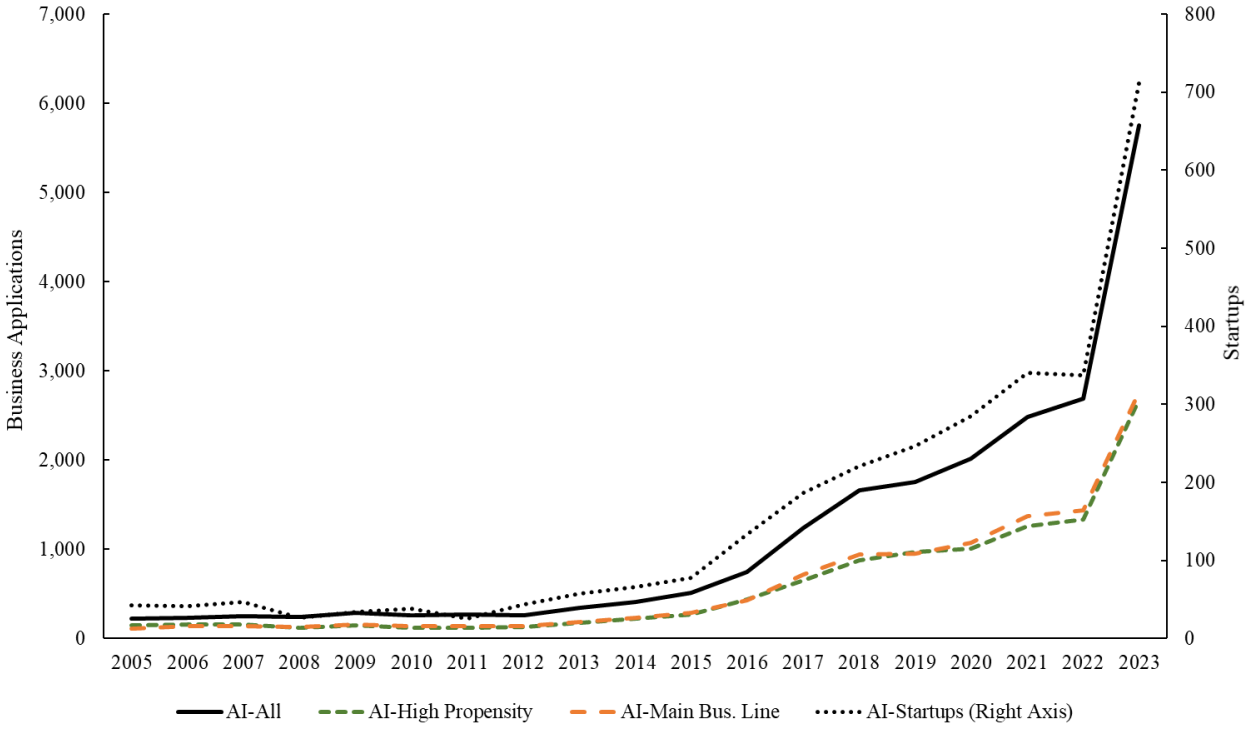
Notes: Includes startups (new employer businesses) from all business applications for the period 2004q3-2021q4. Columns 1-6 provide estimates from an OLS regression conditional on survival – columns 7 and 8 are based on an LPM. The dependent variables are DHS-transformed except for Exit and Transformation, which are indicator variables (e.g. 1 if the business exits in a given year; 0 otherwise). Firm "transformation" in Column 8 takes on a value of 1 if the firm transforms into a different single unit or into a multi-unit firm – see Section 4.7 for the full definition. All continuous variables are winsorized in the right tail at 99.9%. The revenue sample is much smaller because of missing revenue for many firms in the LBD – firm-level weights in LBD based on revenue-reporting propensity were used in regressions involving revenue to make the sample representative. Standard errors clustered at the firm level are in parentheses. (*), (**), and (***) represent significance at 10%, 5%, and 1%, respectively. The independent variables are indicators of: an AI application (ai (all)), planned first wage payment date (wage), application from a corporation (corp), starting a new business or hiring an employee (new bus./hire), existence of a previous EIN (prev. ein), presence of an executive or trade name (exec./trade name), and presence of a business address (bus. address). All regressions include fixed effects for application (BFS) year, state, application week (of the year), industry x LBD year (4-digit NAICS x year). We also control for categories of firm age (measured as LBD year - BFS year). Columns (4)-(8) also control for firm size class. Observation counts (N) are rounded to protect confidentiality.

Table 8: Estimates used in the simulation of AI use rates

	Estimate	Source
Adoption rate in 2023 (a_{2023})	5.5%	BTOS - Bonney et al. (2024)
De-adoption rate in 2024 (d_{2024})	14.5%	BTOS - Bonney et al. (2024)
Adoption rate by incumbents in 2024 (r_{2024})	3.3%	BTOS - Bonney et al. (2024)
Exit rate of adopters in 2024 (χ^a_{2024})	0.0066	Table 7 and BDS
No. of AI startups in 2023 (E^a_{2024})	932	Figure 1
Avg. growth rate of no. of startups (2012-2023)	31%	Figure 1
Total no. of firms in 2024 (N_{2024})	5,467,840	BDS
Avg. growth rate of total no. of firms (g) (2017-2021)	0.56%	BDS

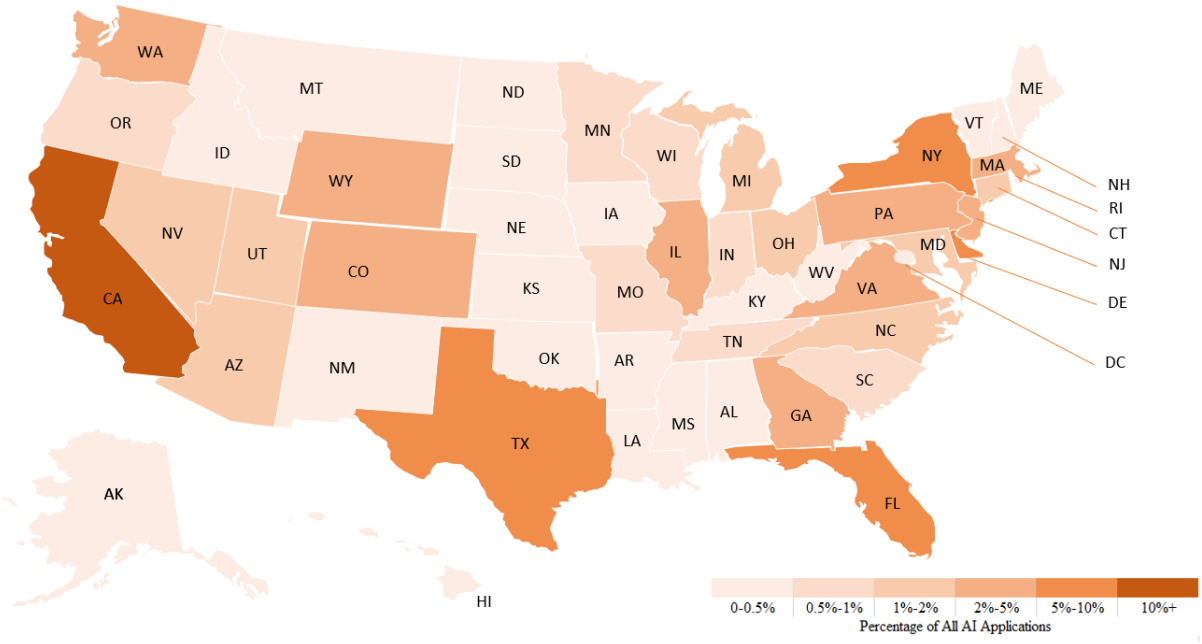
Notes: Growth rate of total no. of firms is the average annual growth rate calculated over the years 2017-2021. This average growth rate also gives an estimate for the total number of firms in 2024, starting in 2021. The annual growth rate in AI startups is calculated for the period 2012-2023 based on Figure 1.

Figure 1: AI business applications and formations per year



Notes: Uses all AI business applications for the period 2005-2023m10, AI business formations (also referred to as transitions or startups) within 8 quarters of application for the period 2005-2019, and predicted AI business formations for the period 2020-2023m10. Plotted are all AI business applications (AI-All), AI business applications based on main business line (AI-Main Bus. Line), AI business applications based on business/trade name (AI-Bus./Trade Name), and AI employer startups originating from applications within 8 quarters of application (AI-Startups) – uses predicted startups after 2019.

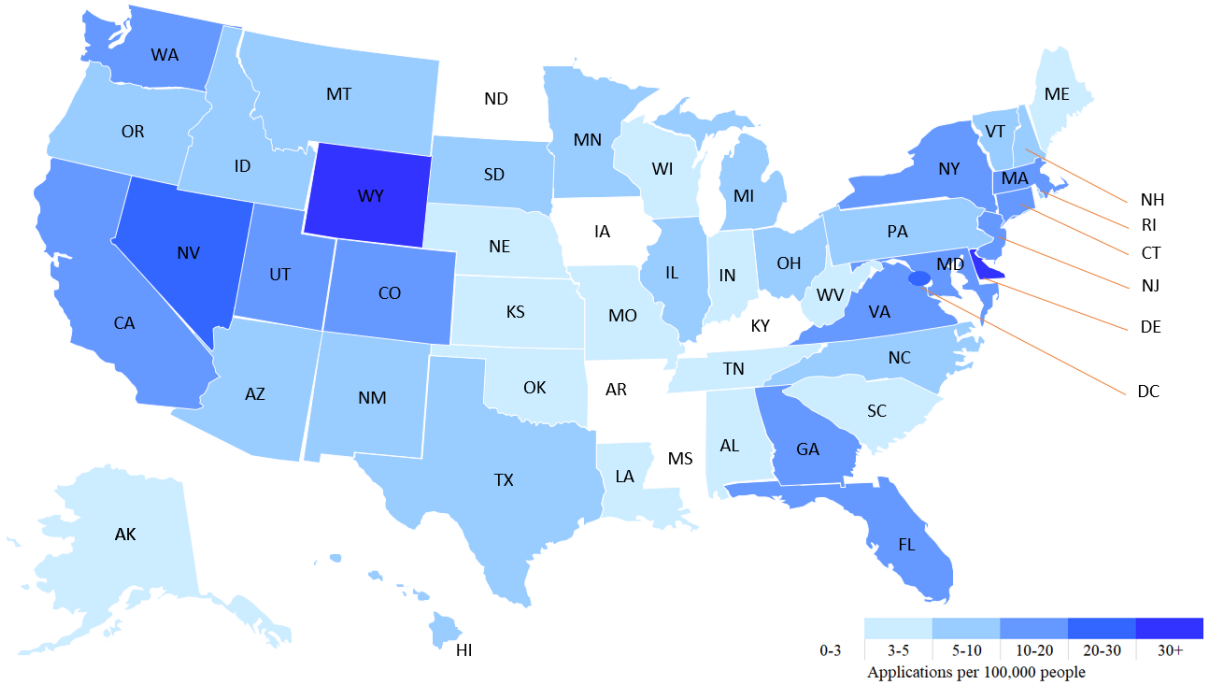
Figure 2: The share (%) of AI business applications by state



Notes: Includes all AI business applications for the period 2004q3-2023m10.

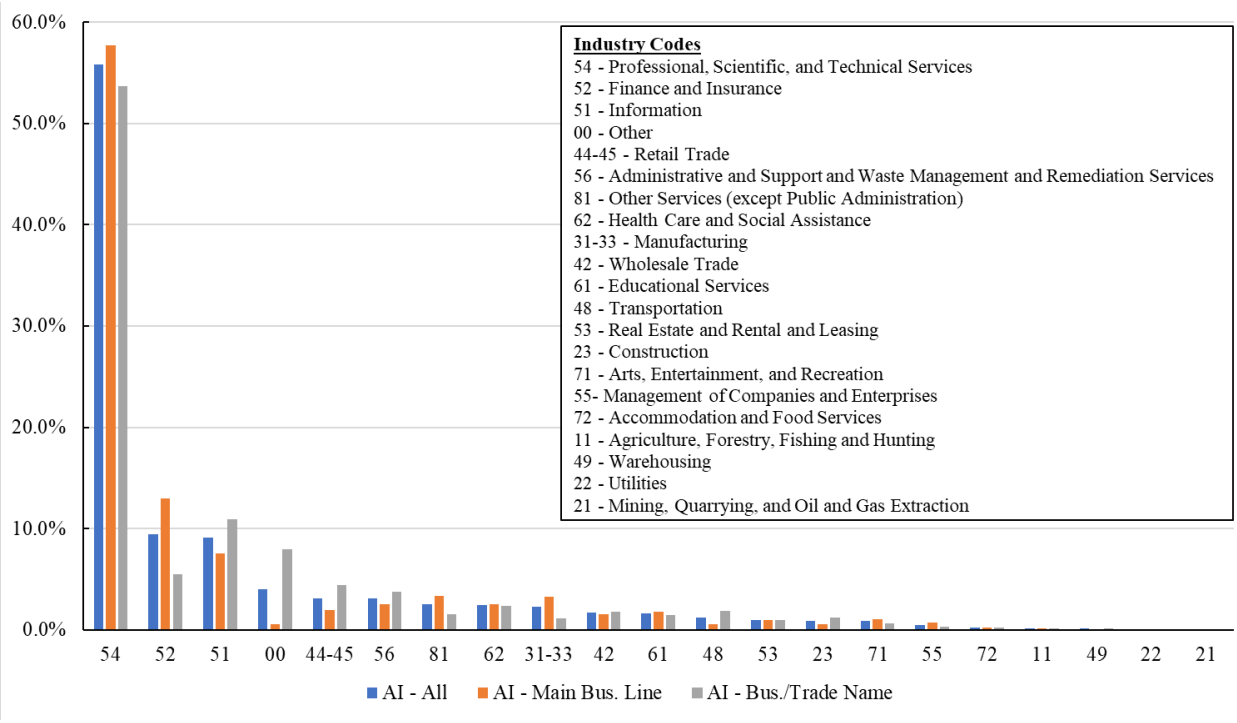
Figure 3: AI business applications per capita by state

(per 100,000 residents of age 18-64)



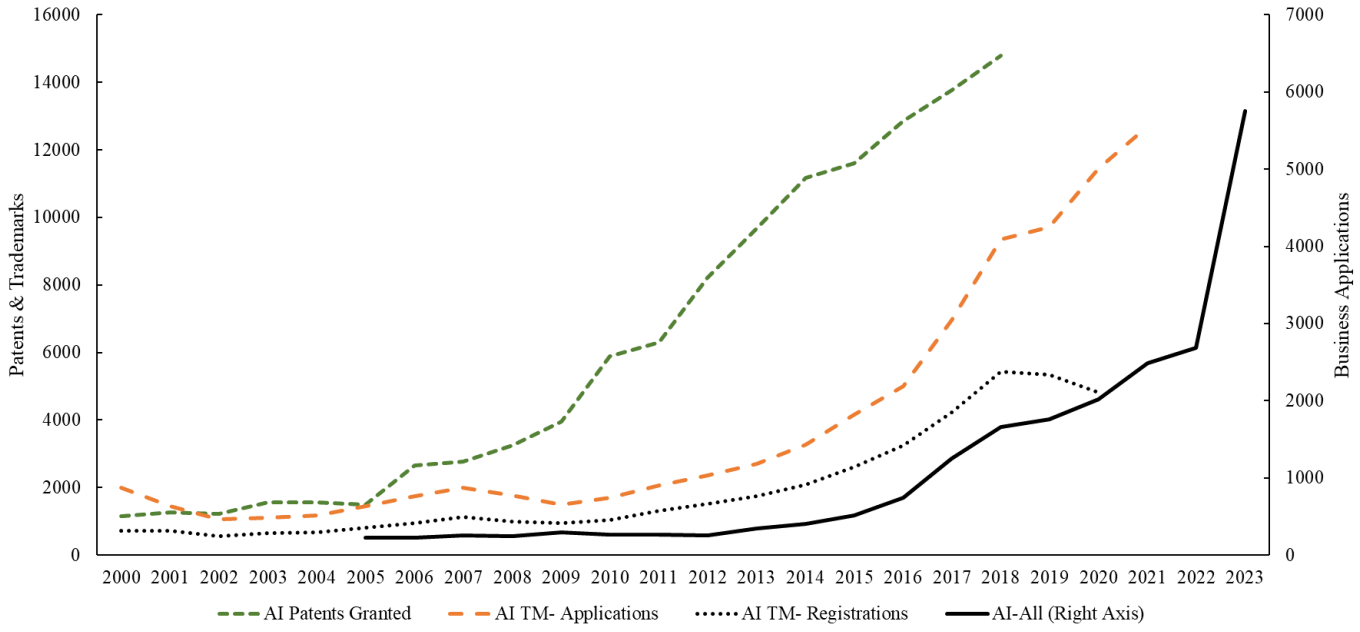
Notes: Includes all AI business applications for the period 2004q3-2023m10. Uses state-level average estimated resident population for ages 18-64 over the period 2005-2022 – estimates are from U.S. Census Bureau.

Figure 4: The share (%) of AI business applications by sector



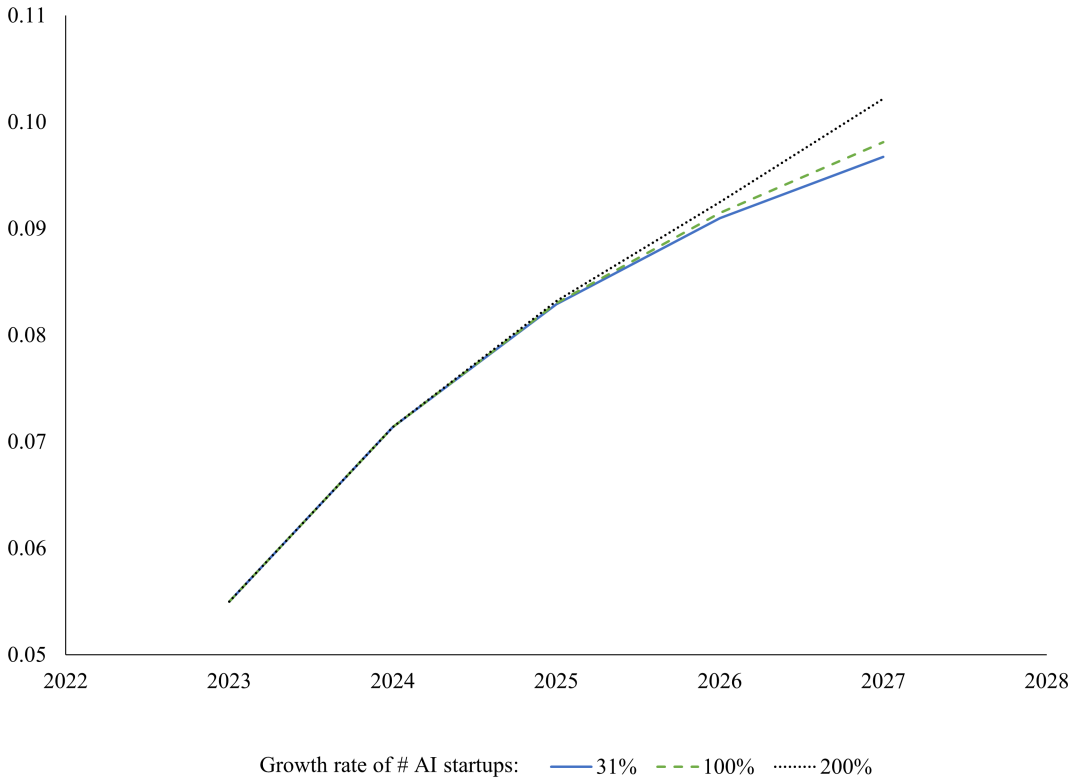
Notes: Includes all AI business applications 2004q3-2023m10. Sectors are in decreasing order of AI business applications (AI-All) share from left to right. Sector code "00" refers to applications that are not classified into any NAICS sector by the industry assignment algorithm used for BFS. AI-Main. Bus Line and AI-Bus./Trade Name shares for NAICS 21 and 22 are suppressed to protect confidentiality.

Figure 5: AI patents, trademarks, and business applications per year



Notes: Includes all AI business applications for the period 2005-2023m10. AI patents are those that are identified by Carnegie Mellon University with 95% confidence. Trademark applications and registrations were identified using text search on United State Patent and Trademark Office's (USPTO) Trademark Case File Database (TCFD).

Figure 6: Estimates of AI Use Rate, 2023-2027

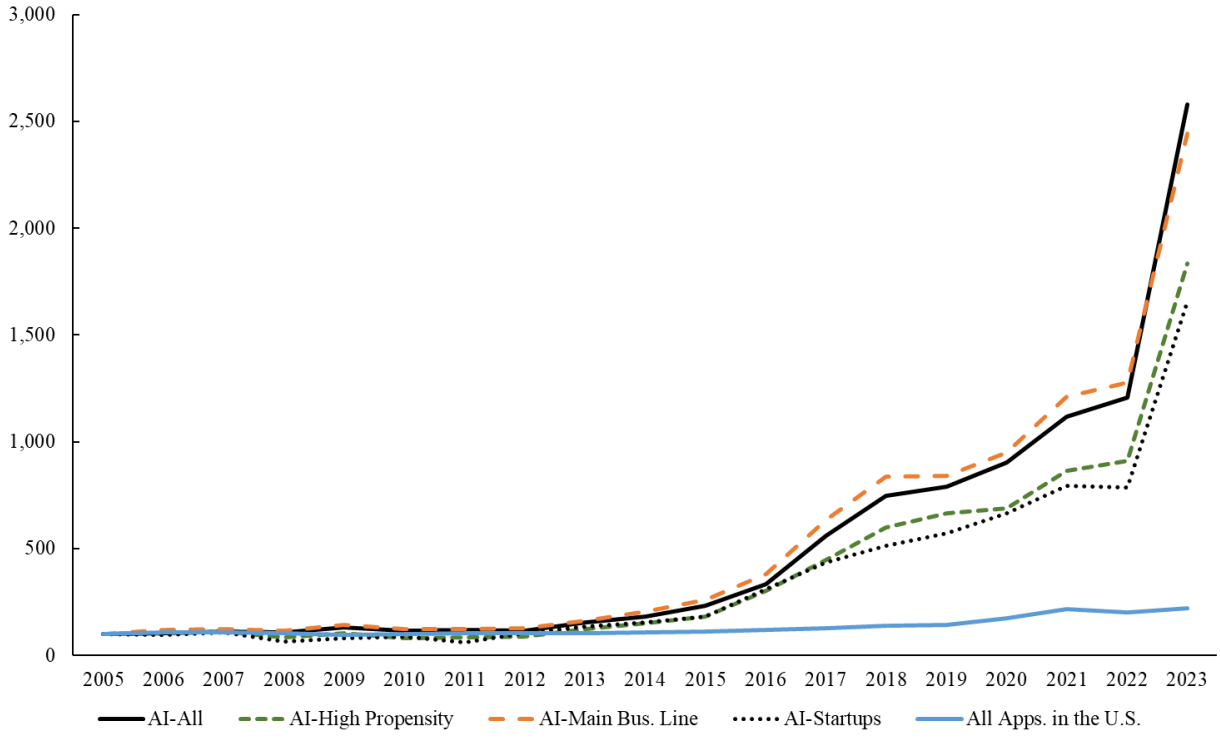


Notes: The figure presents three scenarios for the time-path of AI use rate among firms for the period 2023-2027. The scenarios are based on the law of motion for AI adoption rate, a_t, in 4. Each line corresponds to a different assumption about the annual growth rate of the number of AI startups: 31%, 100%, 200% – see Section 4.8 for details.

Appendix

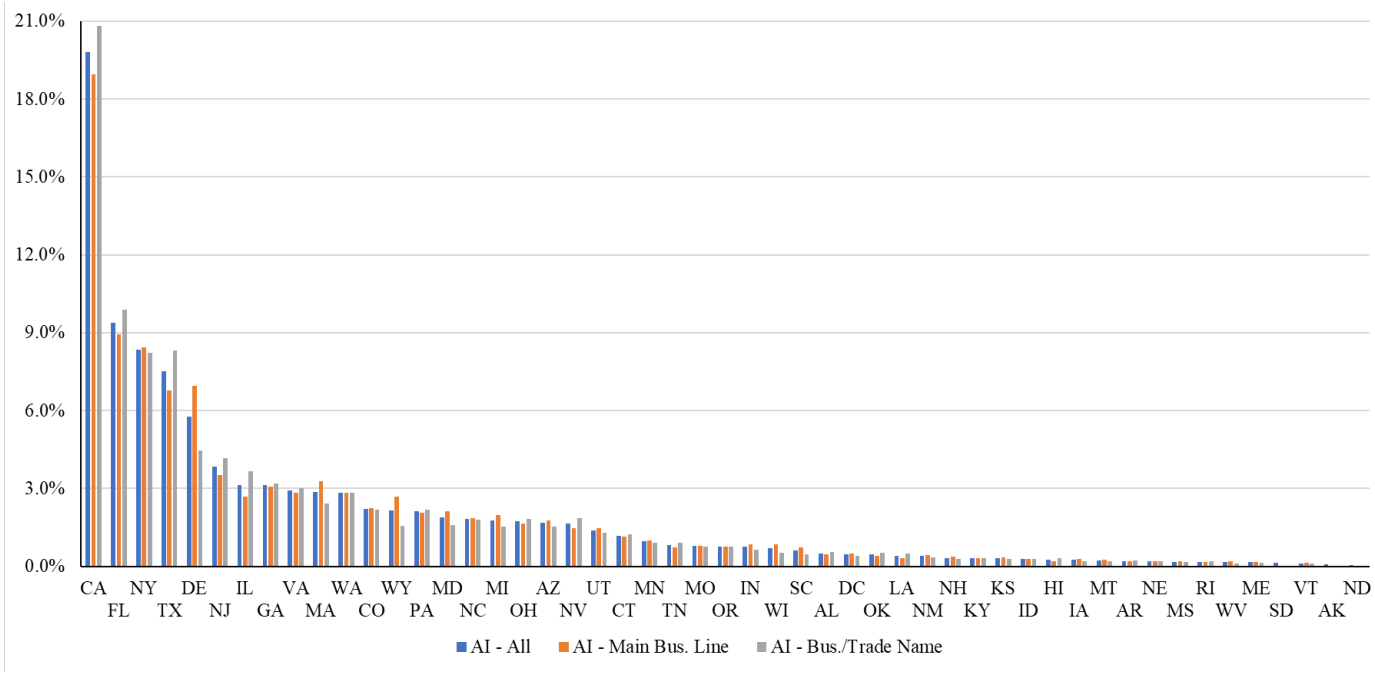
A Additional figures

Figure A.1: AI business applications and formations per year (Index_2005=100)



Notes: Uses all AI business applications for the period 2005-2023m10, AI business formations (also referred to as transitions or startups) within 8 quarters of application for the period 2005-2019, and predicted AI business formations for the period 2020-2023m10. Plotted are all AI business applications (AI-All), AI business applications based on main business line (AI-Main Bus. Line), AI business applications based on business/trade name (AI-Bus./Trade Name), and AI employer startups origination from the applications within 8 quarters of application (AI-Startups) – uses predicted startups after 2019. Each series is indexed to 100 for 2005.

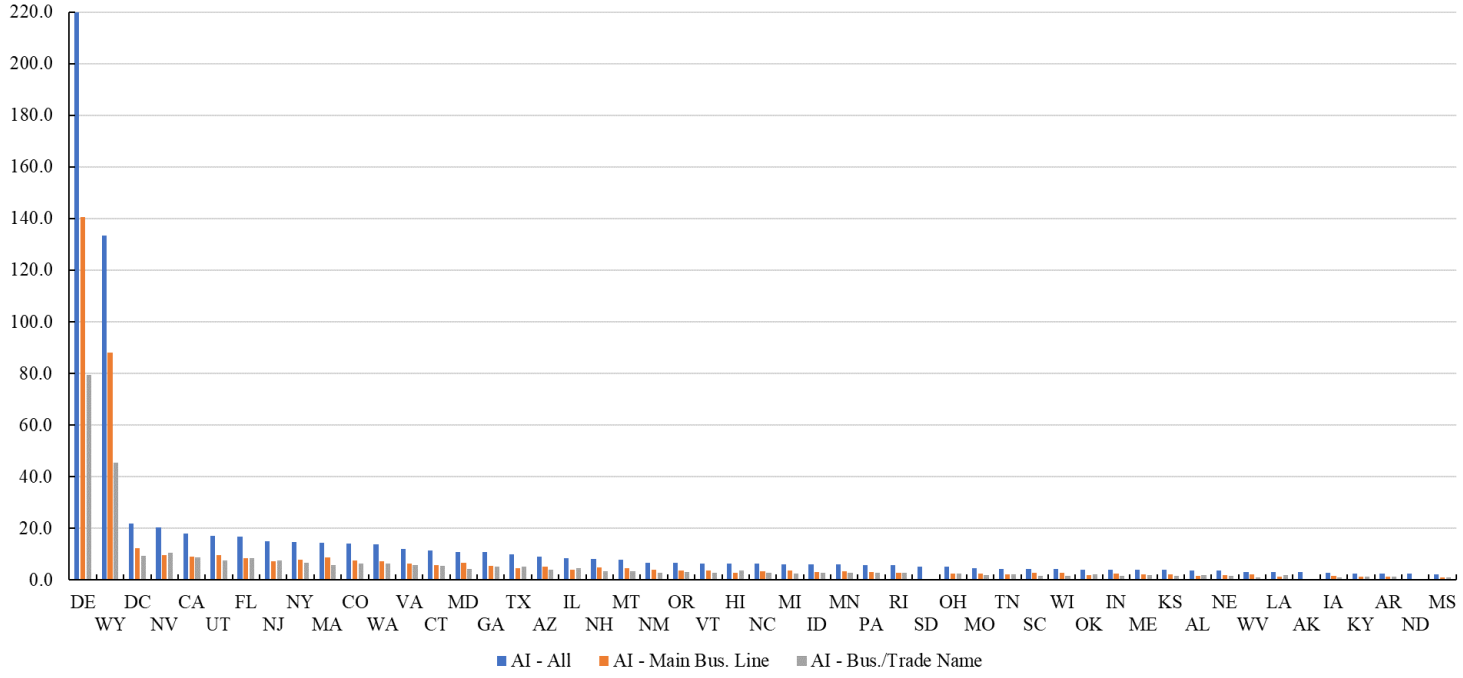
Figure A.2: The share (%) of AI business applications by state



Notes: Includes all AI business applications for the period 2004q3-2023m10. States are in decreasing order of AI business applications (AI-All) share from left to right. AI-Main Bus. Line and AI-Bus./Trade Name shares for AK, ND, and SD are suppressed to protect confidentiality.

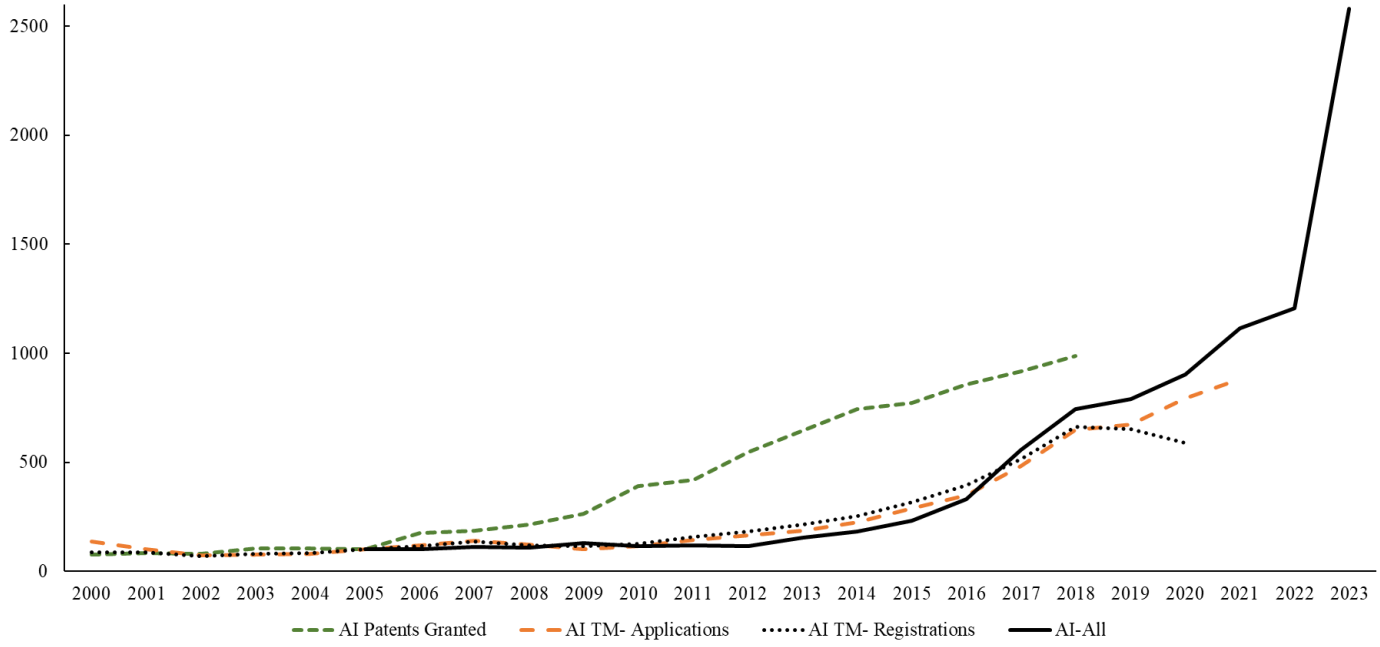
Figure A.3: AI business applications per capita by state

(per 100,000 residents of age 18-64)



Notes: Includes all AI business applications for the period 2004q3-2023m10. Uses state-level average estimated resident population for ages 18-64 over the period 2005-2022 – estimates are from U.S. Census Bureau. States are in decreasing order of AI business applications (AI-All) per capita (per 100,000 residents of age 18-64) from left to right. AI-Main. Bus Line and AI-Bus./Trade Name shares for AK, ND, and SD are suppressed to protect confidentiality.

Figure A.4: AI patents, trademarks, and business applications per year (Index_2005=100)



Notes: All series are indexed to 100 for 2005 to facilitate the comparison of trends with that of AI business applications (AI-All). AI patents are based on the identification by Branstetter et al. (2023). AI trademark applications and registrations are based on text search using USPTO's Trademark Case File Database (TCFD) and AI-related keywords.